Perturbation theory of linear systems with applications to structured epidemic models

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Abstract

We explore the impact of parameter changes on infection spread within structured populations. We study perturbations of linear models in both discrete and continuous time. By interpreting the resulting formulas in the context of an associated Markov chain, new insights are gained. Specifically, we explain how the rate of change of key quantities, that we represent with the so-called canonical flow, is determined by expected hitting times of corresponding states in the Markov chain, from different initial distributions. We give intuitive direct proofs of known results and obtain novel formulas in the case of a general primitive or irreducible Metzler matrix. We apply our results to two models from the literature: the fast pathogen movement limit of a structured model of Cholera spread, and the slow-fast decomposition of critical structured SIR models near the disease-free set. We explain how the latter context can be used to model the effects of control measures with uncontrolled response variables.

Lay Summary

The study of infectious disease necessarily involves a good understanding of how its spread is affected by the underlying conditions of the infection and the network that supports it (contact rates, virulence, etc). In this study, we investigate how changes in parameters related to these underlying conditions affect the spread of infections in populations that are structured in some way, for example by age, location, etc. We obtain some new formulas that describe how these changes occur in the context of an underlying random process that we are able to describe precisely. We give two applications of our methods, one that studies how infection risk changes with respect to pathogen movement rates, and another that describes how infection spreads in a population that is responding to control measures.

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Chapter 1

Introduction

Mathematical modelling is a powerful tool for understanding and predicting the dynamics of infectious diseases. A fundamental mathematical model in this field is the Susceptible-Infectious-Recovered (SIR) model [29], which divides the population into three compartments: susceptible (S), infectious (I), and recovered (R) and is described below. This model is crucial in analyzing the spread of diseases, predicting outbreaks, and formulating effective public health policies. Structured SIR models extend this framework by allowing us to include various demographic and epidemiological factors in the model, which provides a more comprehensive understanding of disease transmission.

For these models, to determine whether a small outbreak can lead to an epidemic, and to understand the dynamics of infection spread when the number of infectious is small, the method to use is known as linearization. Linearizing a model near an equilibrium point means keeping only the linear terms in the equation, as we will see. In the case of disease models, the equilibria of interest are the "disease-free" equilibrium points where there are no infected individuals. Moreover, under a condition known as irreducibility, that is discussed below and defined in Chapter 3, linear models possess three key quantities: the overall growth or decay rate of infection, the disease prevalence (the relative expected number of infected individuals of each type during a large outbreak), and the infection potential (the relative probability that an infected individual of a given type causes a large outbreak). As we will explain, these quantities are supplied by the well-known Perron-Frobenius theorem.

The main goal of this work is to present a new way of understanding how changes in transmission and recovery rates affect the growth rate, prevalence and potential of a linear model (and thus of an infection model linearized near a disease-free equilibrium). Previous work on the so-called perturbation theory of these key quantities has mostly focused on using techniques from linear algebra (a notable exception being the 1975 paper [56], whose work we

extend in a couple of directions as discussed in Chapter 3). These techniques give concise formulas, but can suffer from a lack of intuition. Our approach works by associating a Markov chain (a simple kind of random process, see Chapter 3) to the model, that encapsulates the information that is relevant to the infection spread. Using this perspective, the desired formulas have a simple interpretation in terms of the average time required by the Markov chain to travel from one state to another, called expecting hitting time. We include two applications that relate to structured SIR models: the first is a simpler and more intuitive proof of a result from [38] that computes the change in pathogen distribution, over a finite set of nodes, as a function of pathogen movement rates between nodes. The second is the so-called slow-fast decomposition of the structured SIR model, that we obtain under certain apparently specific, but very natural assumptions: the number of infectious is small relative to the number of susceptibles (small-I), and never becomes too small, or too large, relative to its initial value (criticality). These assumptions make sense, for example, in the setting of an epidemic that persists for a long time, in which the fraction of the population that is infectious at any given time is fairly small, and for which the infection neither dies out completely, nor completely takes over the population.

Before going into greater detail, let us first examine the basic SIR model. The three compartments of this model, namely S, I and R, help us track how individuals move through different disease states over time. Now, let us explore the equations that govern this model:

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I. \end{aligned} \tag{1.1}$$

Here, β is the rate of transmission of the infection and γ is the recovery rate. Both are assumed to be positive. Notice that the total population size $S(t) + I(t) + R(t)$ is constant over time (its time derivative is equal to 0); unless otherwise mentioned we will assume it has been scaled so that $S+I+R = 1$. If the population is initially almost all susceptible, i.e., $S_0 \approx 1$ with I_0 small but positive, then initially $dI/dt \approx (\beta - \gamma)I$, so the infection grows exponentially if $\beta > \gamma$ and decays exponentially if $\beta < \gamma$. These cases can also be expressed in terms of the *basic reproduction number* $R_0 := \beta/\gamma$: the infection grows if $R_0 > 1$ and decays if $R_0 < 1$. It can be shown that

this expression agrees with the standard definition of R_0, which is that R_0 is equal to the total number of infections caused by an infected individual, before that individual recovers, in a completely susceptible population.

The basic SIR model assumes all individuals respond to infection in the same way. Of course, in reality, an individual's response to infection will depend on many factors such as age, genetics, etc. This is where structured models come into play. In a structured model, the population is divided into n groups, giving variables S_j, I_j and R_j for $j = 1, \ldots, n$, with equations

$$\begin{aligned} dS_j/dt &= -\sum_{i=1}^{n} I_i \beta_{ij} S_j \\ dI_j/dt &= \sum_{i=1}^{n} I_i \beta_{ij} S_j - I_j \gamma_j \\ dR_j/dt &= I_j \gamma_j \end{aligned} \tag{1.2}$$

where $\beta_{ij} \geq 0$ is the transmission rate from type i to type j and $\gamma_j > 0$ is the recovery rate of type j. Again, the total population size $\sum_j (S_j + I_j + R_j)$ is constant over time. Let us now see how to linearize a structured SIR model near an equilibrium point.

A solution of (1.2) is constant over time if and only if $I(0) = 0$, so let $(S^*, 0, R^*)$ denote an arbitrary equilibrium point of (1.2). When $I(0)$ is small, the non-linear terms in the equations become relatively less significant compared to the linear terms. This allows us to simplify the analysis by approximating the system's behaviour using linear equations. The general technique to perform linearization involves the matrix of partial derivatives of the right-hand side of the equations in (1.2) with respect to the state variables $S_1, \ldots, S_n, I_1, \ldots, I_n, R_1, \ldots, R_n$. This forms the "linear part" of the equations and is called the Jacobian matrix, and denoted J. The linearized dynamics then takes the form

$$\frac{dy}{dt} = yJ \quad \text{where} \quad y = (S_1, \ldots, S_n, I_1, \ldots, I_n, R_1, \ldots, R_n).$$

Instead of computing the full Jacobian, which is a bit messy, let us focus on just the I dynamics. If $\max_j I_j$ is small, which is true close to any point of the form $(S^*, 0, R^*)$, then S changes slowly, so with the approximation $S(t) \approx S^*$, the linearized equations for dI/dt are given by

$$dI_j/dt = \sum_i I_i \beta_{ij} S_j^* - \gamma_j I_j.$$

Note that the right-hand side is a linear function of the vector $(I_1, \ldots, I_n)$. We can write it down in matrix form as

$$dI/dt = I(\beta \circ S^* - \mathrm{diag}(\gamma)) = IB \tag{1.3}$$

where $B = \beta \circ S^* - \mathrm{diag}(\gamma)$ and $\circ$ denotes the Hadamard (entry-wise) product. The entries of B have the following interpretation: if $j \neq i$ then $B_{ij} = \beta_{ij} S_j^*$ is the rate at which a type i infectious individual infects a susceptible type j individual, and on the diagonal, $B_{ii} = b_i - \gamma_i$ where $b_i = \beta_{ii} S^*$ is the rate at which a type i infects a susceptible type i, and γ_i is the recovery rate of type i. Note that the matrix B in (1.3) is constant so the solution of the linearized system is $I(t) = I(0)e^{Bt}$. It should be noted that the basic reproduction number R_0 can also be defined in this context, using the so-called *next-generation matrix* (which can be computed using the method described in [18]) that measures the same quantity as before, just in matrix form, i.e., for each (i, j), the (i, j) entry of the next-generation matrix is the total number of type j infections caused by an infectious type i individual, before it recovers, in a completely susceptible population. R_0 is then defined to be the spectral radius, i.e., the maximum of the absolute values of the eigenvalues, of the next-generation matrix.

Since the entries of β and S^* are non-negative, the off-diagonal elements of the matrix B are non-negative. This type of matrix is called a Metzler matrix (Definition 3.6). In order for the growth rate, prevalence and potential to be well-defined, it is important that B also has the property of irreducibility (Definition in Chapter 3), i.e., that it is possible for the infection to spread from individuals of any type to individuals of any other type. Equivalently, there are no isolated subpopulations. The relevant result for obtaining these quantities is the Perron-Frobenius theorem, Theorem 3.10, a well-known result that applies to irreducible Metzler matrices and is useful in population modelling and epidemiology. The theorem, in its usual form, is stated in terms of so-called primitive matrices, so we also discuss how it is connected to Metzler matrices in Chapter 3. The Perron-Frobenius theorem introduces the Perron value, denoted r, along with its corresponding left and right Perron vectors, labelled u and v respectively. The Perron value r signifies the overall growth rate of the disease, a vital indicator of its spread within a population. The left Perron vector u is called the "prevalence", since it gives the relative distribution of the number infectious as a function of type. The right Perron vector v is referred to as "infection potential", representing the contribution of different types to the overall potential for

disease transmission. See [19] for an example of their usage.

Before moving onto the chapter summaries, let us first introduce the concept of slow-fast decomposition of the structured SIR model (the precise mathematical formulation is in Chapter 4). When we linearized the model, we assumed that I is small relative to S, which implies that S changes slowly relative to I, and we made the approximation $S(t) \approx S^*$. In other words, when I is small relative to S, the I dynamics are fast relative to the S dynamics, i.e., $I_1, \ldots, I_n$ are the "fast" variables. If the fast variables are well-behaved – for example, if the solutions of the fast equations do not grow or decay exponentially but instead tend towards non-zero equilibrium points – then by viewing the process on a sufficiently long time scale we can observe gradual changes in the S variables, without the I variables behaving erratically. The equations that describe these gradual changes are called the "slow" system. On the slow time scale, the fast dynamics happens instantaneously, so in the limit of time scale separation (in this case, as the number infectious tends to 0), the fast variables are assumed to be at equilibrium at all times, responding instantaneously to changes in the slow variables.

In Chapter 2, we review recent advancements in disease modelling and intervention methods and explain connections to our work. We discuss work that addresses several considerations such as rumours, climate change, and the interplay between symptomatic and asymptomatic behaviour of infections. We also investigate work on several common intervention methods, including quarantine, social distancing, and vaccination, along with their respective limitations and effectiveness in combating diseases. Additionally, we delve into the formation of social contact matrices and their application in studying individual-based models and demographically-structured models. Moreover, we examine some scholarly articles demonstrating the use of perturbation in the contact network to understand the spread of diseases. We also discuss usage of the slow-fast decomposition method to analyze systems with distinct fast and slow variables. We study several papers to understand how this technique is used in disease models. As far as we can tell, the slow-fast formulation that we discuss in Chapter 4 has not yet been considered.

Chapter 3 contains the perturbation formulas and their interpretation. It has three parts. The first part is a discussion of the Perron-Frobenius theorem, the quantities that it gives (growth rate, prevalence and potential), and its use in the relevant models. In the second part we obtain a set of

differential equations (DEs) describing how the Perron vectors vary as the underlying matrix is changed. Since we need to make a specific choice of vectors, and since our choice leads to the simplest form of these DEs, we name the solutions of these equations the "canonical flow" of Perron vectors. We also address existence and uniqueness of solutions to the canonical flow equations. In the third part we associate a Markov chain to a given primitive matrix and relate the change in u and v to the expected hitting times of states in the Markov chain. This allows us to obtain new formulas for the perturbations that have intuitive interpretations. As an application we give an intuitive proof of a result from Kirkland, Shuai et al. [38] that relates to the fast pathogen movement limit of a Cholera model from [21]. The application involves analyzing the change in network risk that occurs at the nodes of a star network when adding new connections between leaf nodes.

In Chapter 4 we show that, in a structured SIR model, if I is small, and if the growth rate of the infection is zero at all times (criticality), then there is a separation of time scales (I dynamics are fast, S and R dynamics are slow) and in the limit we obtain a simplified slow-fast system, that is closely related to the canonical flow equations. We give a numerical algorithm for the limiting system and give some simulations to illustrate its dynamics.

Chapter 5 is the Conclusion.

Chapter 2

Literature review

This chapter has two goals. The first goal is to briefly summarize the existing mathematical theory that is used to obtain the results of Chapter 3. The second goal is to provide an overview of relevant literature in mathematical epidemiology, highlighting the diversity of modelling approaches and some potential applications of our results.

2.1 Theoretical background

In this section we discuss some topics and terminology that are explained in Chapter 3; for the terminology that is not defined here, the reader may consult Chapter 3. The Perron-Frobenius theorem [31] is a cornerstone of linear algebra, introducing Perron values and their associated left and right Perron vectors. In the realm of disease modelling, these vectors play a crucial role in understanding infection dynamics [19]. They encapsulate essential quantities that shed light on how infections spread or recede within populations and the relative impact across demographic groups during outbreaks [19]. A natural question in this setting is whether the Perron values and vectors are differentiable with respect to the underlying matrix, and how to compute their derivatives. In [15] the author develops a formula for second-order derivatives of the Perron root of an irreducible Metzler matrix A, with respect to changes in the entries of A, using group inverses of irreducible M-matrices (a square matrix where all the off-diagonal elements are non-positive and each row sum is non-negative) to investigate the concavity and convexity of the Perron root. The following year, the author developed formulas for the first and second-order derivatives of appropriately normalized Perron vectors with respect to the matrix entries and examined the convexity properties of the Perron vectors [16]. In chapter 3, we use one of these formulas to find the derivative of the Perron value and corresponding left and right Perron vectors u, v using the specific choice of normalization $uv' = 0$. The Perron-Frobenius theorem implies existence of the stationary distribution of an irreducible and aperiodic Markov chain, so it is natural to look for connections to that theory. In [56], the author discusses the role

of the group inverse in the theory of Markov chains. The author introduces a series formula for the group inverse $A^{\#}$ (which is $Q^{\#}$ in our case) of the matrix $A = I - T$ ($Q = I - A/r$ in our case), where T is a stochastic matrix. The author established the defining properties of the group inverse using existing characterizations of the group inverse that were proved in another article. We used the same formula and discussed not only stochastic but also primitive matrices (and their continuous-time analogues) and also proved the defining properties of group inverse directly from the formula.

2.2 Modelling and applications

In this section, we describe some of the relevant literature in mathematical epidemiology. We list the various mathematical modelling approaches that are taken to understand infection spread. We discuss some of the intervention methods that are used to control infection. We explain the meaning and relevance of the contact matrix in the context of structured models, and discuss some work that studies the effect of perturbing the contact matrix. Finally, we discuss the use of slow-fast decompositions to underscore the study of infection models in the presence of different timescales. For each topic, we briefly explain the connection to our work including potential applications, and, when it is relevant, we compare and contrast the literature with our results.

2.2.1 Modelling approaches

Over the years, epidemiological research has proposed many approaches to tackle the spread of infectious diseases. Researchers have examined the problem from different perspectives, exploring various aspects of disease transmission and control, ranging from mathematical models to machine learning algorithms. This diversity of approaches has enabled a more nuanced understanding of the factors that drive disease spread and has contributed to developing more effective interventions to mitigate the effects of outbreaks. Three general categories of approaches have been proposed in [71] for addressing the problem of infectious disease transmission. The first category involves statistical methods for monitoring outbreaks and identifying spatial patterns in real epidemics. The second category consists of mathematical models, also called state-space models, which forecast the spread of an epidemic in a hypothetical or ongoing scenario; for example, the structured SIR model discussed in the Introduction belongs to this category. The third category comprises machine learning and expert methods, which

can be used to forecast the evolution of an ongoing epidemic. By leveraging these approaches, researchers can gain a more comprehensive understanding of the factors driving disease spread and develop more effective strategies for outbreak management. In the Introduction, we saw how the quantities that are studied in Chapter 3, namely, the prevalence and potential, i.e., Perron vectors of the rate matrix, are related to state-space models, by linearizing near disease-free equilibrium points. In principle, however, any approach that uses prevalence and potential to draw conclusions can benefit from the results provided by our theory.

When creating models for infectious diseases, it may be important to account for a wide range of factors that can affect the spread of the disease, such as age, sex, immunodeficiency, climate conditions, and disease-carrying insects. Different model frameworks have been proposed to address various epidemiological questions related to these factors [73]. The most basic such factor is disease status, which is accounted for by so-called compartmental modelling. The basic SIR model discussed in the Introduction is an example of a compartmental model with three compartments, namely, S, I and R. Compartmental modelling is used to model a wide range of infectious diseases, including HIV/AIDS [5, 33], influenza [2], and COVID-19 [89]. A more realistic, but more complicated approach is agent-based modelling, which consists in representing individual agents or actors and their interactions with others in the population. This approach is used to model infectious diseases such as dengue fever [49, 60] and COVID-19 [41, 69]. Agent-based modelling often includes explicit social networks and interactions between individuals, represented as nodes and edges in a graph; some examples are models of sexually transmitted infections [47] and COVID-19 transmission [88]. Structured modelling (for example, the structured SIR model from the Introduction) is a somewhat simpler way to incorporate population structure and heterogeneity into a model. This approach has been used to model infectious diseases in different age groups, such as pertussis in infants [28] and COVID-19 [13, 63]. Each of these modelling approaches offers different insights into the transmission dynamics of infectious diseases, allowing researchers to gain a comprehensive understanding of factors such as population-level disease spread, individual-level interactions and behaviours, and the impact of demographic and spatial heterogeneity on disease transmission. The choice of approach will depend on the specific research question and available data. These models have various applications in biology other than infectious disease modelling [51, 52]. Additionally, these models can be refined to improve their correspondence with

real-world data, providing more accurate insights [34, 82]. The results of Chapters 3 and 4 are applicable to structured models, in the manner discussed in the Introduction, but not directly to agent-based models, since in order for linearization to be valid, the number of susceptibles of each type must be large, requiring the population to be "lumped" into a relatively small number of categories.

Recent research has demonstrated that mathematical models can help to identify the characteristics of diseases that most impact their spread. The authors of [35] have developed a new mathematical model, referred to as the Θ-SEIHRD model, for analyzing the spread of COVID-19. This model considers the disease's distinctive characteristics, such as undetected cases and hospitalized individuals' diverse sanitary and infectious conditions. Notably, the model introduces a novel parameter, denoted as Θ, representing the fraction of detected cases over the total number of infected cases. In [80], authors incorporate early-stage asymptomatic as well as later-stage symptomatic infections for modelling and surveillance of dynamics of infectious diseases. Authors of [32] studied the spread mechanism of rumours on social network platforms during the spread of COVID-19. They also explored using education as a potential control measure for limiting the spread of rumours. Some researchers have explored the concept of herd immunity in their modelling of infectious diseases. A few examples are [8, 43, 44]. Some climate-sensitive diseases consider climate-related parameters to model infectious diseases. These diseases are often influenced by factors such as temperature, humidity, and rainfall [10, 59].

2.2.2 Intervention methods

Mathematical modelling of disease spread primarily aims to identify the most effective intervention strategies, considering various factors such as socialization patterns, governmental capabilities, socio-economic conditions, environmental factors, and more. Mathematical models are now useful tools that policymakers can use to determine intervention strategies during a pandemic [58]. However, the selection of intervention methods can be challenging due to the complex and dynamic nature of disease transmission. Non-pharmaceutical interventions, such as personal hygiene, social distancing, closure, and quarantine measures are the most common intervention methods to slow the spread of illness [23]. Pharmaceutical interventions, such as vaccinations, are one of the most effective ways to prevent the spread of diseases, as they create immunity in individuals and reduce the number of

susceptible individuals in a population [65]. Social distancing aims to reduce contact between individuals and minimize the transmission of infectious diseases. Quarantine measures are also used to control the spread of diseases by isolating individuals who are infected or have been in close contact with infected individuals. These measures have been employed in various outbreaks, including the recent COVID-19 pandemic, and have shown to be effective in controlling the spread of the disease [25, 45, 63, 87].

However, the effectiveness of these intervention methods depends on the specific disease and the population's characteristics. For example, vaccination programs may not be effective for diseases with high mutation rates or low vaccine efficacy, and social distancing and quarantine measures may be difficult to implement in populations with high population density or limited access to resources [42]. Moreover, these measures may have economic, mental and social consequences, such as loss of income, unemployment, and reduced access to education and healthcare services [3]. The results of Chapter 3, which explain the effect of changes in disease parameters on important disease-related quantities, can be used to assess the effects of an intervention on the distribution of the infection, and of the risk of an outbreak, on a population, especially when that intervention corresponds to a small change in parameters.

2.2.3 Contact networks

Social interaction has long been recognized as a key factor in the spread of infectious diseases, and in recent years, it has been increasingly incorporated into disease modelling efforts. The most obvious way to implement a social network model is to represent individuals with nodes and relationships with edges between pairs of nodes. Small community networks can easily be visualized using conventional graphs, but this is not always true for large community networks. However, since people tend to follow some pattern in their socialization, it is sometimes possible to summarize social interactions among large communities by lumping individuals into categories, which amounts to replacing an agent-based model with a structured model. For example, in [61], authors analyzed real social networks and uncovered some intriguing patterns. The research reveals that individuals often connect with others who share similar traits, such as age, race, religious beliefs, or gender. Unsurprisingly, geographical proximity also plays a role, as people tend to prefer forming connections with those who are physically close to them. By defining a set of categories and then determining the level

of connection between individuals in those categories, we obtain a contact matrix. Since the contact matrix influences transmission rates, our theory can be used to understand the effect that changing entries in the contact matrix has on important disease-related quantities. We now discuss some related work that makes use of contact matrices, and also discuss intervention methods related to them.

The structure of contact networks plays a significant role in the spread of infectious diseases in humans [1] and wildlife [84]. The individual variations in social behaviour and interaction patterns influence the emergent properties of these networks, which in turn affect disease dynamics [30]. Network analysis, such as contact tracing, can identify high-risk groups and assess the effectiveness of targeted interventions. Furthermore, wearable devices and location-tracking technology have allowed for more accurate tracking of social interactions, enabling the development of more realistic models of disease spread [24]. To set up an effective intervention method, it can be helpful to first understand contact and population mixing patterns. In [62], authors focus on understanding the social contact patterns and mixing among students in K-12 schools in Pittsburgh, Pennsylvania, USA. The study used data from a school-based survey to construct the social contact network and analyzed the mixing patterns among different grades, genders, and classrooms. The findings of this study can be useful for designing targeted interventions to control infectious disease spread in schools, such as vaccination to certain groups, part of school closure etc.

The authors of [76] explore the effectiveness of different interventions in controlling the spread of SARS-CoV-2, including school closures, workplace closures, and physical distancing measures. The authors also investigate the impact of these interventions on contact patterns between individuals and how this affects the virus's transmissibility. In [63], authors used age-specific reopening and vaccination policy to control COVID-19 spread, with the timing and level of reopening determined by the healthcare system's readiness. As COVID patients are more likely to go to the ICU, the authors considered the availability of ICU beds as an important measure of the healthcare system's readiness. The authors of [53] have proposed an optimal control model incorporating four key control variables: public health awareness and education, vaccination, preventive measures, and intensive care. The authors of [14] discussed when quarantine is an effective intervention method, and here they argued that quarantine could be an effective tool in controlling outbreaks. Still, its effectiveness depends on various factors, such as the

disease's infectiousness, the speed at which it spreads, and the effectiveness of other control measures. The authors state that a well-executed quarantine strategy implemented during an outbreak can potentially mitigate a specific number of infections, and they claim an 80 percent likelihood that this assertion holds true within specified upper and lower bounds of accuracy.

There are several control measures that depend on the type of disease. Here are a few. Article [4] discusses hand hygiene in controlling norovirus spread in nursing homes. In [85], authors investigate the airflow patterns generated by people during various breathing and coughing scenarios, with different types of face covers, to understand their effectiveness in preventing SARS-CoV-2 transmission.

While attempting to control a particular disease through an intervention method, unintended consequences can arise that may worsen another disease. For example, [17] explores the impact of interventions targeting the community network on disease risk. With lockdowns and restrictions in place, many individuals, including those with TB, have likely spent more time at home. As a result, there could be a higher risk of TB transmission among household contacts. So, finding the right intervention is always challenging because the impact of community network interventions on disease risk can be unpredictable and have unintended consequences.

2.2.4 Perturbation of contact networks

During an epidemic, susceptible individuals may actively avoid socializing with infected individuals, which can be modelled using a so-called adaptive network [79] [83]. Perturbing the network structure or intentionally altering node connections can significantly impact disease dynamics. By studying the effect of perturbing different connections it may be possible to identify so-called critical points in the network where targeted interventions can be applied to reduce disease spread. Another challenge is implementing equitable interventions that do not disproportionately affect vulnerable populations. For example, social distancing measures may be difficult to implement in communities where individuals rely on close social interactions for economic or social support. This type of study is one area of research in which our results can be directly applied.

The idea of perturbation has proven useful in areas other than the modelling of infectious diseases. For instance, in the study conducted in [27],

the authors explore how altering social networks can foster the development of self-sufficiency among individuals contending with homelessness and substance use disorders. Through the implementation of a training program aimed at building healthier social networks, positive changes were observed in the lifestyle behaviours of participants. The findings of this study demonstrate the potential utility of perturbation as a tool for driving positive changes in human behaviour and well-being in contexts beyond disease spread. In [81], a physiologically structured population model of T-cell populations at the end of an immune response was investigated using perturbation theory. The authors employed perturbation theory analysis to understand the effects of small changes on the death and absorption rates of physiological variation. They observed that these rates were less pronounced than the average rates, resulting in the division of the heterogeneity into two components: a homogeneous part and a structure dependence. This approach facilitated a more comprehensive understanding of the system's underlying dynamics, particularly the role played by the T-cell population structure in determining the overall immune response. We hold a specific interest in implementing perturbation theory in infectious disease modelling. Therefore it would be advantageous to explore this topic in greater detail.

We are highly motivated by the paper [38], which introduces novel mathematical techniques to investigate how the basic reproduction number R_0 (described in the Introduction) changes when the structure of a network is modified. The approach enables the analysis of two types of model networks: a star network and a path network. The star network is a hub-and-spoke model representing human transport between a central hub, such as a large city, and several spokes representing small cities or suburbs. On the other hand, a path network consists of multiple communities situated sequentially along a track and can represent communities along a river or stream. The flexibility of the frameworks of the star network allows us to use it in modelling various possible scenarios, such as a central water source with several communities supplied by that source or individuals having interactions primarily with a central point that plays a significant role in disease transmission. The researchers employed their findings to explore possible strategies for controlling disease outbreaks by introducing new connections to a network or altering the strength of existing connections. They discovered that the placement of the hot spot(s) (location(s) with a high rate of infection spread) and the connections among patches play a critical role in determining the optimal approach for reducing the risk of infection in both the star and path networks. The proposed mathematical techniques provide

a quantitative assessment of the effectiveness of different strategies in controlling infection. As explained in Section 3.3.3, our results enable a more intuitive understanding of the effect of perturbation in this setting.

While perturbation analysis is widely used in disease models, there has been limited research on perturbing disease-related parameters and analyzing their effects. That said, there are several articles that discuss perturbation in the context of disease spread. For example, the paper [77] investigates the impact of changing case definitions for COVID-19 on mainland China's epidemic curve and transmission parameters. The authors analyze the daily reported cases of COVID-19 and the resulting transmission parameters during different periods of case definition changes. Changing the definition of 'cases' significantly impacts the epidemic curve and transmission parameters. The study highlights the importance of standardized case definitions and data reporting in effectively monitoring and controlling the spread of infectious diseases. The authors of [48] conducted research using Costa Rica as a case study to develop a network-based model of Zika virus (ZIKV) transmission. The model incorporated both mosquito-human and heterosexual transmission dynamics using mean-field theory. Additionally, the authors examined the potential effectiveness of modifying the sexual network structure (i.e., the degree distribution) in controlling the spread of ZIKV in Costa Rica. The results shows that effectively suppressing the spread of Zika virus (ZIKV) at the outset of an epidemic can be achieved by closely monitoring and isolating high-risk contacts and implementing measures such as quarantine and closure of locations with a high concentration of sexual partners to reduce the degree heterogeneity. It is also recommended to employ similar measures to interrupt ZIKV infection and reduce the overall impact at the onset of an outbreak. In [36], authors discussed regular perturbation methods with a regular time scale and then with a 'corrected' time scale to find the evolution of precancerous lesions caused by the human papillomavirus. Authors of [64] describe an HCV infection model and its parameters. They then studied the model's steady states and their stability. Using a perturbation analysis approach, they showed how the shoulder phase arises during antiviral therapy for HCV infection and provided an approximate formula from which one can calculate its length.

Sometimes changing the social structure of the host population has a mixed impact or almost no impact on disease spread. The authors in [54] examine the transmission of *Mycobacterium bovis* to cattle through badgers. Their findings reveal perturbation in the social organization of badgers due

to culling, resulting in both advantageous and disadvantageous impacts on the incidence of bovine tuberculosis in cattle. The study concludes that although badgers are the host of diseases, changing their social structure may not significantly aid disease control.

2.2.5 Slow-fast decomposition

The method of slow-fast decomposition is useful in systems where one of the time scales is much faster than the others, since it simplifies the analysis by assuming that the fast variables adjust instantaneously to changes in the slow variables, reducing one set of differential equations to a simple functional relationship. To use this method, the requirement is to identify the different timescales on which the system evolves. We present several examples of this method in disease modelling, and finish by contrasting it with the results of Chapter 4.

Several compartmental models, with names such as SIR, SIRS, SEIR, MSEIR etc., are typically analyzed using standard tools for ordinary differential equations in order to understand the system's global behaviour. This approach can provide valuable insights into the overall dynamics of the disease, such as the epidemic threshold [11, 20], the final size of the epidemic [22, 78], and the impact of control measures [67, 86]. However, this approach may not capture the complexities of population heterogeneity, such as differences in mixing patterns or vaccination coverage. Standard analysis also falls short in situations where accurate pandemic forecasting requires complex modelling techniques and an in-depth understanding of the biological and epidemiological knowledge of the disease [57]. In contrast, the slow-fast decomposition analysis approach can provide a more detailed understanding of the dynamics of infectious diseases in populations with heterogeneities. By separating the fast and slow variables, this approach can reveal the different timescales on which the infection spreads and the factors that drive the spread on each timescale. In [68], the authors use social awareness as a natural control method for the SIR model. They accomplish this by dividing the susceptible population into the aware and the unaware groups. These two groups switch faster than disease transmission and other biological processes. The findings demonstrate that an increase in the switching ratio and a decrease in infectivity among susceptible populations significantly reduces the disease's prevalence. In [6] authors examine a fundamental model of an epidemic caused by a vector-borne disease. In this model, humans act as hosts and follow a basic SIR model, while mosquitoes act as vectors and

follow a simple SI model where they do not recover from infection. Due to the difference in time scales between hosts (slow) and vectors (fast), the authors use a slow-fast decomposition to determine the final size relations of the epidemic. In [70], authors investigate the dynamics of malaria with relapse using two mathematical models. The human population is divided into four classes: susceptible, exposed, symptomatic, and asymptomatic, while the vector population is divided into susceptible and infectious categories. Now, the average life span of humans is several decades, while the average life span of mosquitos is only several weeks. This statement leads the authors to use two different time scales to analyze the full dynamics of the disease. In the research article, [46], the authors discuss the slow-fast dynamics in the Malaria model caused explicitly by Plasmodium. The study focuses on the transmission of malaria infection between two individuals, namely those with the HbS (Haemoglobin Sickle-cell) gene and those with the normal gene. One significant observation is that individuals with the HbS gene tend to have a smaller population size compared to those with the normal gene, which results in slower population dynamics. This phenomenon introduces the slow-fast dynamics that are associated with malaria infection. The authors of [26] use a so-called singular perturbation method to investigate a model of TB-HIV co-infection. This model is divided into two sub-models, with the HIV sub-model having slower dynamics and the TB sub-model having faster dynamics. The dynamics of the original model can be established and analyzed by studying the reduced models, whether fractional or not. By focusing on the slow model of HIV, the authors provide a comprehensive stability analysis. They find that the reduced slow model associated with its disease-free equilibrium can completely approximate the full TB-HIV co-infection model when the basic reproduction number is less than 1. In the above examples, the slow-fast decomposition occurs because there are two different mechanisms that operate on different time scales. In contrast, in Chapter 4 of this work, the slow-fast decomposition occurs because the number of infectious individuals is small compared to the number of susceptibles, and is stable in the sense of not experiencing either exponential growth, or exponential decay. Although they are formulated for the structured SIR model, the results of Chapter 4 can be applied to any compartmental model in which these assumptions hold.

Chapter 3

Linear models in discrete and continuous time

In the Introduction, we saw that linearizing the structured SIR model near a disease-free state leads to the linear system of differential equations $dI/dt = IB$, where I is the row vector counting the number of infectious individuals of each type, and B, the rate matrix, has the property that all its off-diagonal entries are non-negative, i.e., B is a Metzler matrix (definition below). We can also consider linear models in discrete time (see equation (3.1)), in which case the relevant quantity is the so-called "reproduction matrix" A, with the interpretation that a_{ij} is the expected number of type j infections that are caused by an infected type i individual in one time step. A reproduction matrix has the property that all its entries are non-negative. As discussed for the Metzler matrix case in the Introduction, being irreducible is a useful property in both discrete and continuous-time models.

Before describing the results of this chapter, we need a few definitions.

Definition 3.1 (Non-negative Matrix)**.** A matrix is non-negative if all its entries are non-negative.

Definition 3.2 (Stochastic Matrix)**.** A non-negative matrix is stochastic if its rows sum to 1, i.e., $\sum_j a_{ij} = 1 \ \forall i$ [50].

Definition 3.3 (Primitive Matrix)**.** A non-negative square matrix $A = (a_{ij})$ is said to be *primitive* if there exists $k \in \mathbb{N}$ such that $A^k > 0$, i.e., if there exists $k \in \mathbb{N}$ such that for all i, j, the (i, j) entry of A^k is positive.

Definition 3.4 (Irreducible)**.** An $n \times n$ matrix A is said to be *irreducible* if, for all i, j such that $1 \leq i < j \leq n$, there exists a positive integer k and $i = i_0, i_1, \ldots, i_k = j$ such that $a_{i_0 i_1} a_{i_1 i_2} \cdots a_{i_{k-1} i_k} \neq 0$. Equivalently, defining a graph with vertices $1, \ldots, n$ and a directed edge from i to $j \neq i$ if and only if $a_{ij} \neq 0$, A is irreducible if any vertex can be reached from any other vertex in a finite number of steps.

Definition 3.5 (Aperiodic)**.** A non-negative matrix A is aperiodic if for each i, the period of i, defined by

$$\text{period}(i) = \gcd\{k : (A^k)_{ii} > 0\},$$

is equal to 1.

If A is non-negative, irreducible and aperiodic then it is primitive [74].

Definition 3.6 (Metzler Matrix)**.** A Metzler matrix is a matrix in which all the off-diagonal components are nonnegative, i.e., $\forall_{i \neq j}, a_{ij} \geq 0$.

Note that the main diagonal of a Metzler matrix may contain any values, positive, negative or zero.

Definition 3.7 (Generator Matrix)**.** A generator matrix is a Metzler matrix that has rows that sum to 0, i.e.,$\sum_j a_{ij} = 0 \ \forall i$.

The exponential of a generator matrix is stochastic [55], and the exponential of an irreducible Metzler matrix is primitive (see Lemma 3.11).

Definition 3.8 (Markov Chain)**.** A finite-state discrete time Markov chain $\{X_n : n = 0, 1, 2, \ldots\}$ is a stochastic process that takes finitely many values (called states) such that for each n and each list $i_0, i_1, \ldots, i_n, j$ of states,

$$\mathbb{P}(X_{n+1} = j | X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}, X_n = i_n)$$

$$= \mathbb{P}(X_{n+1} = j | X_n = i_n).$$

A finite-state continuous time Markov chain $(X_t)_{t \in \mathbb{R}_+}$ satisfies the same conditions, except that $0, \ldots, n, n+1$ are replaced with arbitrary times $t_0 < \cdots < t_n < t_{n+1}$.

In this thesis, all Markov chains have finitely many states, so this will be assumed without comment. Note that if A is a stochastic matrix then A defines a Markov chain by letting

$$\mathbb{P}(X_{n+1} = j \mid X_n = i) = a_{ij} \quad \forall i, j, n.$$

For the Markov chains discussed in this chapter, we will use the notation $\mathbb{P}_i$ and $\mathbb{E}_i$ to denote probabilities and expected values, conditioned on $X_0 = i$ (the initial value of the Markov chain is i).

Definition 3.9 (Expected Hitting Time)**.** Let $\tau_j = \inf\{n : X_n = j\}$ be the hitting time of state j in a Markov chain, which is the first time the Markov chain, starting from an initial state i, reaches state j. The expected hitting time $\mathbb{E}_i[\tau_j]$ is defined as the expected number of time steps it takes for the Markov chain to reach state j starting from state i.

Mathematically, in discrete time it can be expressed as:

$$\mathbb{E}_i[\tau_j] = \sum_{n=0}^{\infty} n\, \mathbb{P}_i(\tau_j = n).$$

where $\mathbb{P}_i(\tau_j = n)$ is the probability that τ_j takes on the value n when the Markov chain starts from state i.

We now give a summary of this chapter, which has three parts. In the first part, we discuss the well-known Perron-Frobenius Theorem, Theorem 3.10, which describes the asymptotic behaviour of A^n for primitive A, or equivalently of the discrete-time linear system for $(u(n))_{n=0}^{\infty}$ given by

$$u(n) = u(0)A^n \tag{3.1}$$

for arbitrary non-negative row vectors $u(0)$. It asserts that the spectral radius of A is an eigenvalue, that we call the Perron value, and describes the asymptotic behaviour of A^n for large n in terms of the left and right eigenvectors corresponding to the Perron value, which are called Perron vectors. We will show that the Perron-Frobenius theorem also describes the asymptotics of e^{Bt} for large t when B is an irreducible Metzler matrix. This is very likely a known result but is not the standard formulation of the Perron-Frobenius theorem, so we have included the proof (Lemma 3.11). Using this result we can also study the behaviour of linear systems of the form $(u(t))_{t\geq 0}$ given by the equation below, where B is an irreducible Metzler matrix and $u(0)$ is as above:

$$u(t) = u(0)e^{Bt}. \tag{3.2}$$

The linearized I equations from the Introduction are an example of this type of system. Finally, we introduce an example of an n-state life cycle model in order to understand how the left and right Perron vectors can be interpreted as prevalence and reproductive potential, respectively.

In the second part, we study the perturbation theory of Perron vectors, i.e., we consider a function $A(t)$, where t is a variable parameter and $A(t)$

is primitive for all t, and study how the left and right Perron vectors, denoted $u(t)$ (row vector) and $v(t)$ (column vector) respectively, vary with A. Specifically, we obtain differential equations for u and v that involve a type of matrix known as the group inverse. This is a type of generalized inverse that works for some non-invertible matrices, including those encountered here. In order to obtain these equations, we need a normalization assumption, i.e., an assumption that specifies the length of $u(t)$ and $v(t)$ for each t. This is because the Perron vectors, being eigenvectors, are only determined up to their length. We find that the orthogonality assumption $u'(t)v(t) = u(t)v'(t) = 0$ for all t leads to the simplest form of these equations, so we give the solutions to these equations the title of *canonical flow of Perron vectors*. We also prove the existence and uniqueness of solutions to the canonical flow equations (Theorem 3.17), for which we require a series formula for the group inverse, as well as some properties of this formula. The calculation that gives the derivatives $u'(t)$ and $v'(t)$ is borrowed from [15, 16], although we deviate from their calculations after introducing our choice of normalization, since it leads to a simpler form. The series formula, in the special case of a stochastic matrix, is given in [56], although the same formula works for arbitrary primitive matrices as shown in Lemma 3.14. The choice of normalization, and the existence and uniqueness of the canonical flow, are new.

In the third part, we consider again a (single) primitive matrix A and use a process known as stochasticization to associate a stochastic matrix Λ to A. We use Λ and the Markov chain that it generates in order to obtain a new formula for the group inverse that is expressed in terms of the expected hitting times of this Markov chain (Theorem 3.19). As in the first part, we also prove analogous results in continuous-time (Theorem 3.21), i.e., in the context of (3.2). Using the new formula we can also express perturbations of u and v in these terms, giving us a new perspective on the perturbation theory of Perron vectors. We conclude with an example demonstrating the utility of this method.

There are two places in the thesis where my supervisor, Dr. Eric Foxall, made direct contributions, both occurring in this chapter: the proof of Lemma 3.13, and the end of the proof of Theorem 3.17. These results were more technical than expected.

3.1 The Perron-Frobenius theory

The Perron-Frobenius theorem, Theorem 3.10, is a fundamental result in linear algebra that characterizes the spectral properties of primitive matrices.

Theorem 3.10 ([31])**.** *Let A be a primitive $n \times n$ matrix and let $r = \rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}$. Then $r > 0$, r is an eigenvalue of A with one-dimensional eigenspace, and all other eigenvalues λ of A have $|\lambda| < r$. Let u, v be such that $uA = ru$ and $Av = rv$. Then $u_i, v_i > 0$ for all $i \in \{1, \dots, n\}$. Let $P = vu/uv$. Then*

$$(A/r)^k \to P \quad as \quad k \to \infty.$$

Here, r is called the *Perron value* and the vectors u, v are called the corresponding left and right *Perron vectors* respectively. Following [66], u is a row vector and v is a column vector throughout the work.

In order to use the Perron-Frobenius theorem in the context of the continuous-time system (3.2) we want to show that if B is an irreducible Metzler matrix then e^{Bt} is primitive for all $t > 0$. This result seems to be well-known, but we could not find a proof, so we have included the proof here.

Lemma 3.11. *If B is an irreducible Metzler matrix, then e^{Bt} is positive for all $t > 0$, i.e., $e^{Bt}_{ij} > 0$ for all i, j. In particular, e^{Bt} is primitive $\forall t > 0$.*

Proof. For any $t > 0$ and integer $n \geq 1$, $e^{Bt} = (e^{Bt/n})^n$. Moreover, if M, N are positive then MN is positive since $(MN)_{ij} \geq M_{ii}N_{ij} > 0$ for any i, j, so it suffices to prove the result for small $t > 0$. The Taylor series of the exponential is

$$e^{Bt} = I + Bt + \frac{1}{2!}B^2t^2 + \cdots = \sum_{m=0}^{\infty}(Bt)^m/m!.$$

From the above series, $e^{Bt}_{ii} = 1 + B_{ii}t + O(t^2)$, which is positive if t is small. If $i \neq j$, let $k = \min\{m \geq 1 \colon \prod_{\ell=0}^{m-1} b_{j_\ell j_{\ell+1}} > 0 \text{ for some } i = j_0, j_1, \dots, j_m = j\}$. Irreducibility (see Definition 3.4) implies that k is finite. Moreover, if $i_0, \dots, i_k$ are not all distinct then $b_{i_0 i_1} \cdots b_{i_{k-1} i_k} = 0$, since otherwise k would be smaller. Since $(B^k)_{ij}$ is a sum over such products, all of which are non-negative and at least one of which is positive, it follows that $(B^k)_{ij} > 0$.

Also, by definition of k, $(B^m)_{ij} = 0$ for $m < k$. Then, using the series formula with more terms,

$$(e^{Bt})_{ij} = ((Bt)^k)_{ij}/k! + O(t^{k+1})$$

which is positive for small $t > 0$. We explained above how the result for small $t > 0$ implies the result for all $t > 0$, so the proof is now complete. □

From Lemma 3.11, and the fact that $\rho(e^{Bt}) = e^{\alpha(B)t}$ for any matrix B, we recover the conclusion of the Perron-Frobenius theorem for e^{Bt}.

Corollary 3.12. *If B is an irreducible Metzler matrix, then*

$$e^{-\alpha(B)t}e^{Bt} \to P \quad \text{as } t \to \infty,$$

where $P = vu/uv$, $uB = \alpha u$, $Bv = \alpha v$ and $\alpha = \alpha(B) := \max\{\text{Real}(\lambda) : \lambda \in \sigma(B)\}$, and $\sigma(\cdot)$ is the spectrum.

To better understand my work, it is imperative to comprehend the physical interpretation of the system, the Perron root and the Perron vectors. Linear systems, like those presented by equations (3.1) and (3.2), enable us to comprehend expected population size in branching processes and probability distributions in Markov processes. Depending on the context, the definition of these parameters changes. In the ecological context where types are indexed by i, the left Perron vector u with entries u_i gives the asymptotic type distribution of the population, while v depicts the relative fecundity of different types. In the case of infection spread described by a branching process, the left Perron vector u provides the asymptotic relative number of infectious individuals of different types, which portrays the disease prevalence, while the right Perron vector v portrays the infectious potential, i.e., the relative size or risk of an outbreak caused by an infected individual of a given type. In a primitive stochastic matrix, u represents the stationary distribution and $v_i = 1$ for all i [7, 37]. Therefore, understanding how changes in the matrix influence u and v provides valuable insights into these systems. For example, in the infection spread context, it tells us the effect that changing parameters has on the infectious population's long-term distribution and the relative likelihood that an infected individual of a given type will cause a large outbreak.

The Perron-Frobenius theorem establishes that the spectral radius of a primitive matrix is positive and is an eigenvalue with a one-dimensional eigenspace. It also shows that the other eigenvalues have absolute values

strictly smaller than the spectral radius and provides information about the associated eigenvectors. Additionally, the theorem gives a convergence result for the powers of A. When we rescale the matrix A, we obtain a matrix $B = A/r$ that satisfies $uB = u$. The Perron-Frobenius theorem implies that as $k \to \infty$, the powers B^k converge to a matrix P. If A is a primitive stochastic matrix, then the left Perron vector u that satisfies $\sum_i u_i = 1$ is the stationary distribution of the Markov chain, which means that the limiting fraction of time spent in state i is proportional to u_i, and P_{ij} is the limiting probability that a Markov chain associated with A starting in state i will be in state j. If A is a general primitive matrix, P_{ij} is the relative proportion of offspring that are of type j if the population starts with a single individual of type i and evolves for a long time.

The following example shows how, in a relatively simple structured population model, it is possible to interpret the Perron vectors as prevalence and reproductive potential by appealing to a kind of life cycle analysis.

Example 3.1 (Multi-stage model). We are considering a discrete-time life cycle model consisting of n states, from egg (state 1) to adult (state n), where survival probability and reproduction govern the transitions between states. In this model, individuals of type $1, \cdots, (n-1)$ either die or move to the next stage and individuals of type n can reproduce, producing on average R offspring per time step, and may die at each time step. The probability of remaining in each state after one step is denoted by $p_{11}, p_{22}, \cdots p_{nn}$, while the transition probability from one state to the next is represented by $p_{12}, p_{23}, \cdots p_{(n-1)n}$. Specifically, p_{12} denotes the probability of transition from state 1 to state 2, and so on.

To understand how the Perron-Frobenius theorem applies in this example, we begin by constructing the reproduction matrix A from Figure 3.1.

$$A = \begin{pmatrix} p_{11} & p_{12} & 0 & \cdots & 0 & 0 \\ 0 & p_{22} & p_{23} & \cdots & 0 & 0 \\ 0 & 0 & p_{33} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & p_{(n-1)(n-1)} & p_{(n-1)n} \\ R & 0 & 0 & \cdots & 0 & p_{nn} \end{pmatrix}.$$

For a reproduction matrix, a_{ij} is the number of type j produced by a

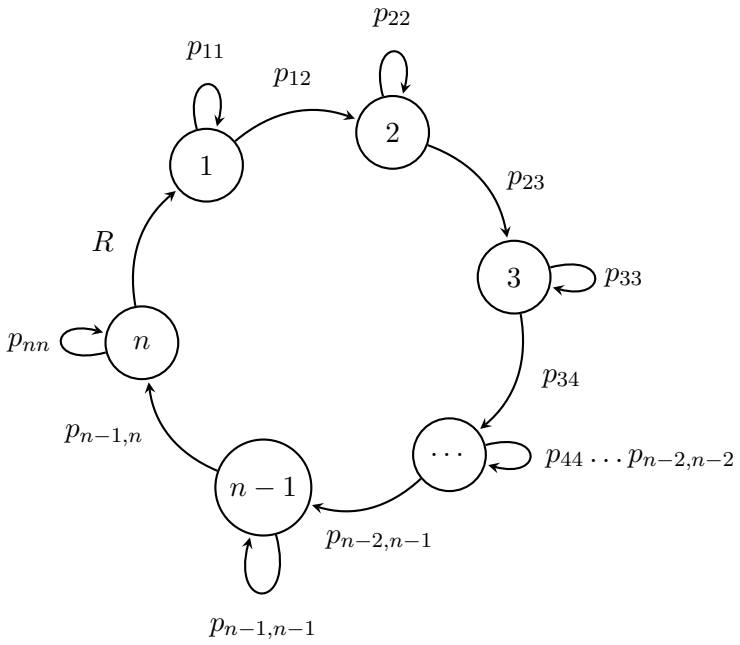


Figure 3.1: Transition diagram of the n-state life-cycle model.

type i in one step. So, $a_{ij} = p_{ij}$ for $j = i$ and $j = i + 1$ and $a_{n,1} = R$. Since $R \geq 0$ and probabilities of transitions and survivals are also non-negative, A is non-negative. As long as $p_{12}, p_{23}, \ldots, R$ are positive, A is irreducible. Our goal for this example is to show that there is a way of interpreting the Perron vectors u and v using a notion of prevalence and reproductive potential, as follows.

First, rescale the matrix A by its Perron value r so that A/r has spectral radius equal to 1 i.e. each egg will produce 1 egg, on average, over its lifespan. So the new reproduction matrix will be in the form:

$$A/r = \begin{pmatrix} p_{11}/r & p_{12}/r & 0 & \cdots & 0 & 0 \\ 0 & p_{22}/r & p_{23}/r & \cdots & 0 & 0 \\ 0 & 0 & p_{33}/r & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & p_{(n-1)(n-1)}/r & p_{(n-1)n}/r \\ R/r & 0 & 0 & \cdots & 0 & p_{nn}/r \end{pmatrix}.$$

We will interpret this in the same way as before, just with all values scaled

by $1/r$, so that the probability to go from 1 to 2 at each step is p_{12}/r, adults produce R/r offspring at each step, etc. The transition probabilities out of states $i < n$ may no longer add up to 1 but as long as $r > 1$ we can view the missing probability as corresponding to death.

We want to show that v represents reproductive potential, or equivalently, relative fecundity, i.e., the expected number of eggs produced by an individual of type i, and that u_i represents the average amount of time spent in state i by an individual that was initially in the egg state (this is how we are interpreting prevalence in the life cycle context).

Let q_i denote the probability, with respect to the rescaled matrix A/r, of transitioning to state $i+1$ before death, given that you are presently in state i. Then we have the following formula for q_i:

$$\begin{aligned} q_i &= \mathbb{P}(i \to i+1 \text{ before dead}) \\ &= \frac{p_{i,i+1}}{r}(1 + p_{ii}/r + p_{ii}^2/r + \cdots) \\ &= \frac{p_{i,i+1}/r}{1 - p_{ii}/r}, \end{aligned}$$

where $1 \leq i \leq n-1$.
To explain: the expression $1 + p_{ii}/r + p_{ii}^2/r + \cdots$ is an infinite geometric series representing the probability of remaining in state i for any number of time steps. Therefore, the expression $(p_{i,i+1}/r)(1 + p_{ii}/r + p_{ii}^2/r + \cdots)$ represents the probability of transitioning from state i to state $i+1$ before eventually leaving state i. The series converges to $(p_{i,i+1}/r)/(1 - p_{ii}/r)$, as long as $p_{ii}/r < 1$.

Since reproduction occurs only in the adult state n, we define q_n to be the expected number of eggs produced by an adult before death:

$$\begin{aligned} q_n &= \mathbb{E}[\#\text{ of eggs produced before dead}] \\ &= \frac{R/r}{1 - p_{nn}/r}. \end{aligned}$$

With these definitions, the expected number of eggs produced by an individual of type i with respect to the rescaled matrix A/r is equal to $q_i q_{i+1} \cdots q_{n-1} q_n$. The first thing we should check is that $\prod_{i=1}^{n} q_i = 1$, i.e., after rescaling, one egg produces on average one egg in its lifetime. To do so, use the fact that r is an eigenvalue of A, then $\det(I - A/r) = 0$. To

find the determinant we can use the Leibniz formula [75]: for any matrix M, with entries m_{ij},

$$\det(M) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} m_{\sigma(i),i},$$

where S_n is the set of all permutations of $\{1, 2, \cdots, n\}$ and $\operatorname{sgn}(\sigma)$ is the sign of the permutation σ. The $\operatorname{sgn}(\sigma)$ of the permutation is determined by $(-1)^{(\text{number of inversions})}$. All the off-diagonal 0 entries in A are also equal to 0 in $I - A/r$, so only two terms in $\det(I - A/r)$ are non-zero: more precisely, if $M = I - A/r$ then $\prod_{i=1}^{n} m_{\sigma(i),i} = 0$ unless σ is the identity or the cyclic shift $\sigma(i) = i + 1 \mod n$. The sign of the cyclic shift is $(-1)^{(n-1)}$ and the sign of the identity is 1. Keeping those two terms and giving them the appropriate signs,

$$\begin{aligned}
\det(I - A/r) &= \prod_{i=1}^{n}(1 - (p_{ii}/r)) + (-1)^{n-1}(R/r) \prod_{i=1}^{n-1}(-(p_{i,i+1}/r)) \\
&= \prod_{i=1}^{n}(1 - (p_{ii}/r)) + (-1)^{2n-1}(R/r) \prod_{i=1}^{n-1}(p_{i,i+1}/r) \\
&= \prod_{i=1}^{n}(1 - (p_{ii}/r)) - (R/r) \prod_{i=1}^{n-1}(p_{i,i+1}/r).
\end{aligned}$$

Since this is equal to 0,

$$\begin{aligned}
\prod_{i=1}^{n}(1 - (p_{ii}/r)) &= (R/r) \prod_{i=1}^{n-1}(p_{i,i+1}/r) \\
\therefore \left(\frac{R/r}{1 - p_{nn}/r}\right) \prod_{i=1}^{n-1} \left(\frac{p_{i,i+1}/r}{1 - p_{ii}/r}\right) &= 1.
\end{aligned}$$

But this is the same as $\prod_{i=1}^{n} q_i = 1$.

Next, we examine the right Perron vector. From the above discussion, we want to show that if we define

$$v_i = q_i q_{i+1} \cdots q_{n-1} q_n, \tag{3.3}$$

where $1 \leq i \leq (n-1)$, then v is a right Perron vector. From our previous computation we know that $v_1 = \prod_i q_i = 1$. To answer this question, let

us approach it from the other side and see what the eigenvalue relation $(A/r)v = v$ gives: for any v,

$$(A/r)v = \begin{pmatrix} p_{11}/r & p_{12}/r & 0 & \cdots & 0 & 0 \\ 0 & p_{22}/r & p_{23}/r & \cdots & 0 & 0 \\ 0 & 0 & p_{33}/r & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & p_{(n-1)(n-1)}/r & p_{(n-1)n}/r \\ R/r & 0 & 0 & \cdots & 0 & p_{nn}/r \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \cdots \\ v_{n-1} \\ v_n \end{pmatrix}$$

$$= \begin{pmatrix} p_{11}v_1/r + p_{12}v_2/r \\ p_{22}v_2/r + p_{23}v_3/r \\ \cdots \\ p_{(n-1)(n-1)}v_{n-1}/r + p_{(n-1)n}v_n/r \\ Rv_1/r + p_{nn}v_n/r \end{pmatrix}.$$

So, if $(A/r)v = v$ then

$$\begin{aligned} p_{11}v_1/r + p_{12}v_2/r &= v_1 \\ p_{22}v_2/r + p_{23}v_3/r &= v_2 \\ \cdots\cdots\cdots & \\ p_{(n-1)(n-1)}v_{n-1}/r + p_{(n-1)n}v_n/r &= v_{n-1} \\ Rv_1/r + p_{nn}v_n/r &= v_n. \end{aligned}$$

Solving we get

$$v_i = \prod_{i=1}^{n-1} \left(\frac{p_{i,i+1}/r}{1 - p_{ii}/r} \right) \left(\frac{R/r}{1 - p_{nn}/r} \right) v_1.$$

Our suggested choice (3.3) has $v_1 = 1$, so (3.3) gives a right Perron vector as long as it satisfies, for all other i,

$$\begin{aligned} v_i &= \prod_{i=1}^{n-1} \left(\frac{p_{i,i+1}/r}{1 - p_{ii}/r} \right) \left(\frac{R/r}{1 - p_{nn}/r} \right) \\ &= q_i q_{i+1} \cdots q_{n-1} q_n, \end{aligned}$$

which it does.

Finally, we examine the left Perron vector. We said that prevalence is represented by the average amount of time spent in state i by an individual that was initially in the egg state. So, if we want to show that if we take

$$u_i = q_{i-1} \cdots q_1 \cdot 1/(1 - p_{ii}/r) \tag{3.4}$$

then u is a left Perron vector of A. Again, we use the eigenvalue relation $u(A/r) = u$ to see what is required.

$$\begin{aligned} u(A/r) &= \begin{pmatrix} u_1 & u_2 & u_3 \cdots & u_{n-1} & u_n \end{pmatrix} \begin{pmatrix} p_{11}/r & p_{12}/r & \cdots & 0 & 0 \\ 0 & p_{22}/r & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{p_{(n-1)(n-1)}}{r} & \frac{p_{(n-1)n}}{r} \\ R/r & 0 & \cdots & 0 & p_{nn}/r \end{pmatrix} \\ &= (p_{11}u_1/r + Ru_n/r \quad p_{12}u_1/r + p_{22}u_2/r \quad \cdots \; p_{(n-1)n}u_{n-1}/r + p_{nn}u_n/r). \end{aligned}$$

Setting this equal to u we get the system of equations

$$u_i = \frac{p_{(i-1),i}/r}{1 - p_{ii}/r} u_{i-1}.$$

Solving recursively,

$$u_i = \frac{p_{(i-1),i}/r}{1 - p_{ii}/r} \cdot \frac{p_{(i-2),(i-1)}/r}{1 - p_{(i-1)(i-1)}/r} \cdots \frac{p_{12}/r}{1 - p_{22}/r} u_1.$$

Our suggested choice (3.4) has $u_1 = \frac{1}{1-p_{11}/r}$ (the expected time spent in the egg state). So, (3.4) gives a left Perron vector if it satisfies, for every $i > 1$,

$$\begin{aligned} u_i &= \frac{p_{(i-1),i}/r}{1 - p_{ii}/r} \cdot \frac{p_{(i-2),(i-1)}/r}{1 - p_{(i-1)(i-1)}/r} \cdots \frac{p_{12}/r}{1 - p_{22}/r} \cdot \frac{1}{1 - p_{11}} \\ &= \frac{p_{(i-1),i}/r}{1 - p_{(i-1)(i-1)}/r} \cdot \frac{p_{(i-2),(i-1)}/r}{1 - p_{(i-2)(i-2)}/r} \cdots \frac{p_{12}/r}{1 - p_{11}/r} \cdot \frac{1}{1 - p_{ii}/r} \\ &= q_{i-1}q_{i-2} \cdots q_1 \cdot \frac{1}{1 - p_{ii}/r}, \end{aligned}$$

which it does.

3.2 Perturbation theory: canonical Flow of Perron Vectors

In this section, we assume that $A(t)$ is a matrix-valued function defined for t in some interval (t_1, t_2), that the entries $a_{ij}(t)$ have continuous derivatives with respect to t and that for all $t \in (t_1, t_2)$, $A(t)$ is a primitive matrix.

We want to compute the derivative of the Perron value r, left Perron vector u and right Perron vector v with respect to t. The following result proves the smoothness of $r(t)$. Dr. Eric Foxall, my thesis supervisor provided the following proof.

Lemma 3.13. *Let $A(t)$ be a matrix-valued function whose entries are C^k functions of t (i.e., have k continuous derivatives) where $k \geq 1$ is an integer. Then, the Perron value $r(t)$ is also a C^k function of t.*

Proof. Consider the function $F(t, \lambda) = \det(A(t) - \lambda I)$, which is equal to 0 if and only if $\lambda(t)$ is an eigenvalue of $A(t)$. By the Perron-Frobenius theorem, for each t, the polynomial $\lambda \mapsto F(t, \lambda)$ has a simple root at $r(t)$ (since $r(t)$ is an eigenvalue with one-dimensional eigenspace), so in particular $\partial F(t, r(t))/\partial\lambda \neq 0$ for each t. By the implicit function theorem [40] there exists a locally unique C^k function $\lambda(t)$ such that $F(t, \lambda(t)) = 0$ and $\lambda(0) = r(0)$. By uniqueness, $\lambda(t) = r(t)$ locally for all t, thus globally for all t, establishing that r is C^k. □

The smoothness of $u(t)$ and $v(t)$, for the given normalization, will follow implicitly from the canonical flow equations.

3.2.1 Derivative of the Perron root

Let A be a primitive matrix with Perron value r, and let u and v be the left and right Perron vectors respectively such that $Av = rv$ and $uA = ru$. Since $uAv = ruv$, perturbation gives

$$u'Av + uA'v + uAv' = r'uv + r(uv)'$$

$$u'rv + uA'v + ruv' = r(uv)' + r'uv$$

$$uA'v + r(uv)' = r(uv)' + r'uv$$

so cancelling $r(uv)'$, we find that

$$r' = \frac{uA'v}{uv}.$$

where uv can be viewed as the total potential of the stationary distribution u and $uA'v$ as the change in potential due to the perturbation in A.

3.2.2 Derivative of the Perron vectors, canonical flow

Suppose we have smooth (continuously differentiable) functions $u(t), v(t)$ that are left and right Perron vectors of $A(t)$ for each t. Then $A(t)v(t) = r(t)v(t)$ for all t. Let $Q(t) = I - A(t)/r(t)$. So

$$\begin{aligned}(I - A(t)/r(t))v(t) &= 0\\ Q(t)v(t) &= 0.\end{aligned}$$

Taking the derivative with respect to (w.r.t) t we get

$$\begin{aligned}Q'(t)v(t) + Q(t)v'(t) &= 0\\ Q(t)v'(t) &= -Q'(t)v(t).\end{aligned} \tag{3.5}$$

Since $Q(t)v(t) = 0$, $Q(t)$ is not invertible. But $Q = I - A/r$ has a group inverse [39] $Q^{\#}$, i.e., a matrix that commutes with Q and behaves like the inverse of Q on the eigenspaces of non-zero eigenvalues, which can be captured by the following three properties: $QQ^{\#} = Q^{\#}Q$, $QQ^{\#}Q = Q$ and $Q^{\#}QQ^{\#} = Q^{\#}$ [39]. Of course, existence is not the same as having a formula. As we show in Theorem 3.16, with $P = vu/uv$ as in the Perron-Frobenius theorem, the following gives a matrix with those three properties:

$$Q^{\#} = (I - (A/r)(I - P))^{-1}(I - P). \tag{3.6}$$

In words, $Q^{\#}$ is obtained by first zeroing out the 1-eigenspace of A/r by multiplying it by $I - P$, putting the modified matrix $(A/r)(I - P)$ into the formula for Q, inverting that, then zeroing out again. Formula 3.6 is proved for stochastic matrices in [56] (in that reference it takes the series form (3.14) discussed below). Essentially the same approach also works for primitive matrices, as we now describe.

Before we prove the $Q^{\#}$ formula we need a few properties of A and P.

Property 3.1. The matrix $P = vu/uv$ is idempotent i.e. $P^2 = P$. Proof: $P^2 = (vu/uv)(vu/uv) = v(uv)u/(uv)^2 = vu/uv = P$.

Property 3.2. If P is idempotent, $I - P$ is also idempotent. Proof:

$$(I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P.$$

Property 3.3. A and P commute, since

$$AP = \frac{Avu}{uv} = \frac{rvu}{uv} = \frac{vru}{uv} = \frac{vuA}{uv} = PA.$$

Property 3.4.

$$(A/r)P = \frac{Avu}{ruv} = \frac{rvu}{ruv} = P.$$

Property 3.5. Since $(I-P)v = v - Pv = v - \frac{vuv}{uv} = v - v = 0$,

$$Q^{\#}v = (I - (A/r)(I-P))^{-1}(I-P)v = 0$$

and similarly $uQ^{\#} = 0$.

3.2.2.1 Validation of $Q^{\#}$ formula

There are a couple of ways to compute $Q^{\#}$. In [38] group inverses are used to investigate the impact of network structure changes on the so-called network risk. We will also use the group inverse method to find the group inverse of $Q = I - A/r$ but will take a simpler and more intuitive route to the desired formulas than the one given in [38]. There are also other types of pseudo-inverses applicable to non-invertible matrix, e.g. Moore-Penrose Inverse, Drazin Inverse etc. But the group inverse is better suited to this context [39]. Since A has eigenvalue r corresponding to eigenvectors u and v and $Qv = uQ = 0$, Q is singular (not invertible). If we remove the top eigenvalue part (i.e., take the matrix that is inverted in (3.6)), then it will be invertible.

Our first task is to show the inverse in (3.6) exists, so let $\Omega = (A/r)(I-P)$. To proceed with the computation, it is necessary first to prove $\rho(\Omega) < 1$ where $\rho(\Omega)$ is the spectral radius of Ω. Once this has been established, the geometric series generated by Ω converges to $\sum_{n\to 0}^{\infty} \Omega^n = (I-\Omega)^{-1}$ [31].

Lemma 3.14. *Let $\Omega = (A/r)(I-P)$ where A is a primitive matrix, r is the Perron value and $P = vu/uv$ where u, v are left, right Perron vectors of A. Then $\rho(\Omega) < 1$.*

Proof. Since A is primitive, A/r has Jordan form SJS^{-1} with

$$J = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & K \end{array}\right),$$

where K has $\rho(K) < 1$ (since, by the Perron Frobenious theorem, 1 is the spectral radius and a simple eigenvalue of A/r). In these coordinates, it is clear that $P = STS^{-1}$ with block form

$$T = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & 0 \end{array}\right)$$

and since $\Omega = (A/r)(I-P) = A/r - P$, $\Omega = SCS^{-1}$ where

$$C = J - T = \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & K \end{array}\right)$$

which has $\rho(C) = \rho(K) < 1$. Since Ω and C have the same eigenvalues, $\rho(\Omega) = \rho(C) < 1$. □

Now to validate our $Q^{\#}$ formula we need to prove our $Q^{\#}$ equation satisfies the three properties of group inverse mentioned earlier. For this task, as well as for later computations, the key fact that makes this true is that $QQ^{\#} = Q^{\#}Q = I - P$, as we show now.

Lemma 3.15. *Let $Q = I - (A/r)$ and $Q^{\#} = (I - (A/r)(I-P))^{-1}(I-P)$. Then $QQ^{\#} = Q^{\#}Q = I - P$.*

Proof. Let $M = I - (A/r)(I-P)$, so that $Q^{\#} = M^{-1}(I-P)$. By Property 3.3, A and P commute, so M and M^{-1} commute with A and P, and Q and $Q^{\#}$ commute, so we only need to show that $QQ^{\#} = I - P$. Using Property 3.4 twice, $(I - A/r)(I-P) = I - A/r - P + (A/r)P = I - A/r$ and $M = I - (A/r)(I-P) = I - A/r + P$, so $I - A/r = M - P$ then using these and commutativity,

$$\begin{aligned} QQ^{\#} &= (I - A/r)M^{-1}(I-P) \\ &= (I - A/r)(I-P)M^{-1} \\ &= (I - A/r)M^{-1} \\ &= (M-P)M^{-1} \\ &= I - PM^{-1}. \end{aligned} \tag{3.7}$$

We need to prove that $PM^{-1} = P$. Using commutativity and the above formula for M,

$$\begin{aligned} P &= MPM^{-1} \\ &= (I - A/r + P)PM^{-1} \\ &= PM^{-1} - (A/r)PM^{-1} + P^2M^{-1}. \end{aligned} \tag{3.8}$$

Since

$$(A/r)P = P \text{ and } P^2 = P,$$

(3.8) becomes

$$PM^{-1} - PM^{-1} + PM^{-1} = P \text{ which implies } PM^{-1} = P.$$

Hence (3.7) becomes $QQ^{\#} = I - P$.

□

Theorem 3.16. *Let $Q^{\#}$ be given by equation* (3.6)*, then*

$$QQ^{\#} = Q^{\#}Q, \quad QQ^{\#}Q = Q \quad \textit{and} \quad Q^{\#}QQ^{\#} = Q^{\#}.$$

Proof. By Lemma 3.15, $QQ^{\#} = Q^{\#}Q = I - P$. So it remains to prove that $QQ^{\#}Q = Q$ and $Q^{\#}QQ^{\#} = Q^{\#}$.

Since $PQ = P(I - A/r) = P - (A/r)P = P - P = 0$,

$$QQ^{\#}Q = (I - P)Q = Q - PQ = Q.$$

Letting $M = I - (A/r)(I - P)$, then using Property 3.2,

$$Q^{\#}P = M^{-1}(I - P)P = M^{-1}(P - P^2) = 0.$$

So,

$$Q^{\#}QQ^{\#} = Q^{\#}(I - P) = Q^{\#} - Q^{\#}P = Q^{\#}$$

and the proof is finished. □

Now, equipped with Lemma 3.15, we multiply both sides of (3.5) on the left by $Q^{\#}(t)$ and, suppressing the t-dependence, we have

$$\begin{aligned} Q^{\#}Qv' &= -Q^{\#}Q'v \\ (I - P)v' &= -Q^{\#}Q'v \\ v' - Pv' &= -Q^{\#}Q'v. \end{aligned}$$

If $uv' = 0$ then $Pv' = (vu/uv)v' = v(uv'/uv) = 0$. Then the v' equation has the tidy form

$$v' = -Q^{\#}Q'v. \tag{3.9}$$

If we do further calculations, we can write the relation as

$$\begin{aligned} v' &= Q^{\#}\left(\frac{A}{r}\right)' v \\ &= Q^{\#}\left(\frac{A'}{r} - \frac{Ar'}{r^2}\right) v \\ &= Q^{\#}\left(\frac{A'}{r}\right) v - Q^{\#}\left(\frac{Ar'}{r^2}\right) v. \end{aligned}$$

Since $Av = rv$ and $Q^{\#}v = 0$ (Property 3.5), we get

$$v' = Q^{\#}\left(\frac{A'}{r}\right)v. \tag{3.10}$$

Similarly, if $u'v = 0$, $\forall t$ then the left Perron vector has

$$u' = -uQ'Q^{\#} = u\left(\frac{A'}{r}\right)Q^{\#}. \tag{3.11}$$

The assumptions $uv' = u'v = 0$ correspond to a specific normalization of the Perron vectors that we call the *canonical flow*.

The following theorem proves the existence and uniqueness of the solutions of the ordinary differential equations that govern the canonical flow.

Theorem 3.17. *Let $t \mapsto A(t)$ be a continuously differentiable matrix-valued function defined for t in some interval (t_1, t_2) containing 0, with Perron values $r(t)$. For any non-zero vectors v_0, u_0 with $A(0)v_0 = r(0)v_0$ and $u_0A(0) = r(0)u_0$ the initial value problems*

$$\begin{cases} v' = -Q^{\#}Q'v, \\ v(0) = v_0 \end{cases} \tag{3.12}$$

and

$$\begin{cases} u' = -uQ'Q^{\#} \\ u(0) = u_0 \end{cases} \tag{3.13}$$

have unique solutions satisfying

$$A(t)v(t) = r(t)v(t), \quad u(t)A(t) = r(t)u(t) \quad \textit{and} \quad u(t)v(t) = u_0v_0$$

for every $t \in (t_1, t_2)$.

Proof. We begin by showing that if v solves the above equations then $A(t)v(t) = r(t)v(t)$ for all t. We have

$$v'(t) = -Q^{\#}(t)Q'(t)v(t).$$

Multiplying on the left by $Q(t)$,

$$Q(t)v'(t) = -Q(t)Q^{\#}(t)Q'(t)v(t).$$

Using Lemma 3.15,

$$\begin{aligned}
Q(t)v'(t) &= -(I-P)Q'(t)v(t) \\
Q(t)v'(t) &= -Q'(t)v(t) + PQ'(t)v(t) \\
&= -Q'(t)v(t) + \frac{v(t)u(t)}{u(t)v(t)}(I - A(t)/r(t))'v(t) \\
&= -Q'(t)v(t) + \frac{v(t)u(t)}{u(t)v(t)}\left(-A'(t)/r(t) + \frac{A(t)r'(t)}{r^2(t)}\right)v(t).
\end{aligned}$$

We'll show the second term is zero: using $uA = ru$ and the relation $r' = uA'v/uv$ from Section 3.2.1 (and suppressing (t)),

$$\begin{aligned}
\frac{vu}{uv}\left(-A'/r + \frac{Ar'}{r^2}\right)v &= \frac{v}{uvr}(-uA'v + uAr'v/r) \\
&= \frac{v}{uvr}(-uA'v + ur'v) = 0.
\end{aligned}$$

Therefore

$$\begin{aligned}
Q(t)v'(t) &= -Q'(t)v(t) \\
\therefore (Q(t)v(t))' &= 0.
\end{aligned}$$

We get the initial condition for v: $A(0)v(0) = r(0)v(0)$. So we can write $Q(0)v(0) = 0$, which gives

$$\begin{aligned}
Q(t)v(t) &= Q(0)v(0) = 0 \\
(I - A(t)/r(t))v(t) &= 0 \\
A(t)v(t) &= r(t)v(t).
\end{aligned}$$

The proof that $u(t)A(t) = r(t)u(t)$ for all t is analogous, so we omit it. Showing that $u(t)v(t) = u_0v_0$ is not hard: since $u(I-P) = (I-P)v = 0$, from the equation for $Q^\#$ it follows that $Q^\#v = uQ^\# = 0$, so it follows from the equations for u' and v' that $u'v = 0$ and $uv' = 0$. It follows from the product rule that $(uv)' = 0$, so $u(t)v(t)$ is constant in t.

Next we prove the existence and uniqueness of the solutions. We treat both equations (3.12) and (3.13) as a single coupled system to be solved. The Picard existence theorem [12] requires continuous dependence in time and locally Lipschitz dependence in the state variables u and v. To understand which terms depend on u, v and which are external inputs, we first give formulas for $Q^\#$ and Q':

$$Q^\# = (I - (A/r)(I-P))^{-1}(I-P) = (I - A/r + P)^{-1}(I-P)$$

and

$$Q' = -(A/r)'.$$

By Lemma 3.13, r is a C^1 (continuously differentiable) function of A, which itself is a C^1 function of t, so Q' is a continuous function of t. Since $u(t)v(t)$ is constant in t for solutions u, v, we may assume $P = vu/c$ where $c = u_0 v_0 \neq 0$. So, P, being a polynomial in u and v, is a locally Lipschitz function of u and v. The remainder of the proof turned out to be more technical than expected, so it was supplied by my supervisor Dr. Eric Foxall.

Matrix inversion, i.e., M maps to M^{-1}, is locally Lipschitz on its domain consisting of all matrices M with $\det(M) \neq 0$. To see this, note the determinant is a polynomial function of the entries and the cofactors are determinants of submatrices, then Cramer's rule gives the inverse as a rational function of the entries whose denominator is zero exactly when the matrix is not invertible (and rational functions are locally Lipschitz on their domain). The function $(u, v) \to I - A/r + vu/c$ is locally Lipschitz in u and v. Thus for each t, $Q^{\#}(u, v, t)$ is locally Lipschitz with respect to u and v, on the set where $\det(I - A/r + vu/c) \neq 0$.

Now, applying Picard's existence theorem [12] to the initial value problem given by equations (3.12) and (3.13), there exists a unique solution $(u(t), v(t))$ defined for t in some interval $(-\epsilon, \epsilon)$ satisfying $(u(0), v(0)) = (u_0, v_0)$. Let (T_1, T_2) denote the largest interval containing 0 on which a unique solution exists. The standard continuation method, in this context, implies that $\det(I - A(t)/r + v(t)u(t)/u_0 v_0) \to 0$ as $t \to T_2^-$, analogously for T_1. The properties we proved above imply that for $t \in (T_1, T_2)$, $u_0 v_0 = u(t)v(t)$ and $u(t), v(t)$ are left/right Perron vectors of $A(t)$. In other words, $\det(I - \Omega(t)) \to 0$ as $t \to T_2^-$ where $\Omega(t)$ is defined for $t \in (T_1, T_2)$ from $A(t)$ as in Lemma 3.14 (note $A/r - P = (A/r)(I - P)$ from Property 3.4). The proof of that lemma shows the spectra are related by $\sigma(\Omega(t)) = \sigma(A(t))/r \cup \{0\} \setminus \{1\}$. Since $t \mapsto A(t)$ is continuous on (T_1, T_2) and eigenvalues of a matrix are continuous w.r.t its entries, if $T_2 < t_2$ then $\lim_{t \to T_2^-} \sigma(\Omega(t)) = \sigma(A(T_2))/r \cup \{0\} \setminus \{1\}$ and in particular $\lim_{t \to T_2^-} \rho(\Omega(t)) = \rho(\Omega(T_2)) < 1$ (defining $\Omega(T_2)$ directly from $A(T_2)$ as in Lemma 3.14, rather than from the solution of the ODEs). Since eigenvalues of $I - \Omega(t)$ are at least $1 - \rho(\Omega(t))$ in magnitude, this contradicts $\det(I - \Omega(t)) \to 0$ as $t \to T_2^-$, so $T_2 = t_2$. The analogous argument show that $T_1 = t_1$.

□

The following theorem will prove the characterization of the canonical flow. It is expressed for the right Perron vector; an analogous result holds for the left Perron vector.

Theorem 3.18 (Characterization of the canonical flow)**.** *Let $A(t)$ be a primitive matrix-valued C^1 function with Perron values $r(t)$ and any left Perron vectors $u(t)$, and let $Q(t) = I - A(t)/r(t)$. Suppose $v : [0,T] \to \mathcal{R}^n$ is a C^1 function satisfying the initial condition $A(0)v(0) = r(0)v(0)$. Then the following are equivalent:*

1. $\forall t \in [0,T]$,
   (a) $A(t)v(t) = r(t)v(t)$ *and*
   (b) $u(t)v'(t) = 0$.
2. $\forall t \in [0,T]$, $v'(t) = -Q^{\#}(t)Q'(t)v(t)$.

Proof. **(1) to (2) :**

The computation of the derivative of the right Perron vector v (see equation (3.9)) proves this part.

(2) to (1) :

(2) to 1(a) is already proved in the first part of the Theorem 3.17. To prove 1(b), starting from the equation in (2), multiply by $u(t)$ on the left and use $u(t)Q^{\#}(t) = 0$ to find that

$$u(t)v'(t) = -u(t)Q^{\#}(t)Q'(t)v(t) = 0.$$

□

In the following section, we will explore the effect that varying the entries in the reproduction/rate matrix has on the corresponding canonical flow vectors u and v. To do so, we first obtain a new formula for $Q^{\#}$ that is expressed in terms of expected hitting times of an associated Markov chain. In the stochastic case, this chain is generated by the matrix A. In the general case, it relates to the interpretation of the model as a branching process of infection spread. In this interpretation, the Markov chain represents, in some sense, the most likely path of an outbreak when the model is normalized to have a Perron value of 1. We will examine both the discrete and continuous time cases and demonstrate how our formula and its properties show up in each context.

3.3 Group inverse and hitting time formula

In this section, we give a novel formula for $Q^{\#}$ expressed in terms of hitting times of an associated Markov chain. This in turn will lead to a formula for perturbations of the Perron vectors (i.e., u' and v', in the notation of the previous section), expressed in the same terms.

3.3.1 Discrete time perturbation analysis

We will cover both the case where A is a primitive stochastic matrix, as well as the general case of a primitive matrix. When A is stochastic, the associated Markov chain is the one with transition matrix A. In the case of a general primitive matrix, the corresponding Markov chain is the one with transition matrix $D^{-1}(A/r)D$, where D is the diagonal matrix with entries $d_{ii} = v_i$, and v is a right Perron vector of A. Note that multiplying v by a non-zero constant does not change $D^{-1}(A/r)D$. This matrix is the *stochasticization* of A. It has entries $(a_{ij}/r)v_j/v_i$, so it is not hard to see that it is stochastic: since $Av = rv$,

$$\sum_j (a_{ij}/r)v_j/v_i = \frac{1}{rv_i}\sum_j a_{ij}v_j = rv_i/(rv_i) = 1.$$

For the Markov chains discussed in this section, we will use the notation $\mathbb{P}_i$ and $\mathbb{E}_i$ to denote probabilities and expected values, conditioned on $X_0 = i$ (the initial value of the Markov chain is i). We will also use $\mathbb{1}$ for the indicator function, i.e., for an event F, $\mathbb{1}(F)$ is the random variable that is equal to 1 for outcomes that belong to F and is equal to 0 otherwise. In particular, $\mathbb{P}(F) = \mathbb{E}[\mathbb{1}(F)]$ for each F.

3.3.1.1 $Q^{\#}$ formula in discrete time:

Since Lemma 3.14 is true the geometric series generated by Ω converges to $\sum_{n\to 0}^{\infty} \Omega^n = (I - \Omega)^{-1}$. We can write (3.6) as:

$$Q^{\#} = \sum_{n=0}^{\infty} \Omega^n (I - P).$$

Since $(A/r)P = P$ (which also implies A, P commute) and since $(I-P)^n = (I-P)$ $\forall n$, $\Omega^n(I-P) = (A/r)^n(I-P)^{n+1} = (A/r)^n(I-P) = (A/r)^n - P$ for $n \geq 0$.

$$\therefore Q^{\#} = \sum_{n=0}^{\infty}((A/r)^n - P). \tag{3.14}$$

This formula appears in [56] in the case where A is a stochastic matrix. The formula (3.14) can be understood as the sum of immediate value minus limiting value. Individual entries are given by

$$Q^{\#}_{ij} = \sum_{n=0}^{\infty}((A^n)_{ij}/r^n) - v_i u_j/uv).$$

If the normalization $uv = 1$ is chosen, this simplifies somewhat to

$$Q^{\#}_{ij} = \sum_{n=0}^{\infty}((A^n)_{ij}/r^n) - v_i u_j).$$

We can consider two cases to interpret the result in the expected hitting time context. If A is a stochastic matrix, then $r = 1$ and for the corresponding Markov chain (X_n), $(A^n)_{ij} = \mathbb{P}_i(X_n = j)$. Moreover, u is the stationary distribution, which we denote by π, and taking $v_i = 1$ for all i gives a right Perron vector that satisfies $\pi v = 1$. Thus, we have

$$Q^{\#}_{ij} = \sum_{n\geq 0}(\mathbb{P}_i(X_n = j) - \pi_j).$$

Otherwise, if A is any other non-negative, irreducible matrix, we need to do the stochasticization of A. To do that let us define $\Lambda_{ij} = (A_{ij}/r)v_j/v_i$. Then $(I - \Lambda)^{\#} = Q^{\#}_{ij} v_j/v_i$ and has the formula

$$(I - \Lambda)^{\#}_{ij} = \sum_{m\geq 0}((\Lambda^m)_{ij} - \pi_j) \tag{3.15}$$

where $\pi_j = v_j u_j$. We have already shown above that Λ is stochastic (has row sums equal to 1), using $Av = rv$.

Let (Y_n) be a Markov chain with transition probability Λ, and let π be the unique stationary distribution for Y. $\mathbb{P}_y(\cdots)$ denotes probability with respect to (Y_n) assuming $Y_0 = y$, and $\mathbb{P}_\pi(\cdots)$ denotes probability assuming Y_0 is distributed according to π. Let $\tau_j = \inf\{n\colon Y_n = j\}$ be the hitting time of state j, the first time the Markov chain reaches state j.

The following result shows that we can understand the entries of the group inverse in terms of hitting times of the underlying Markov chain. This makes specific properties apparent. In [56, Corollary 3.2], the author proved a specific part which is equivalent to the equation $\tilde{Q}^{\#}_{jj} = \pi_j \mathbb{E}_\pi[\tau_j] > 0$ of our theorem. The author used a previously proved mean passage time taken from [72] whereas we prove it directly using the Markov chain.

Theorem 3.19. *Let Λ be a primitive stochastic matrix with stationary distribution π and let $\tilde{Q} = I - \Lambda$. Then*

$$\tilde{Q}^{\#}_{ij} = \pi_j(\mathbb{E}_\pi[\tau_j] - \mathbb{E}_i[\tau_j]). \tag{3.16}$$

In particular,

$$\tilde{Q}^{\#}_{jj} = \pi_j \mathbb{E}_\pi[\tau_j] > 0$$

and for all $i \neq j$,

$$\tilde{Q}^{\#}_{jj} - \tilde{Q}^{\#}_{ij} = \pi_j \mathbb{E}_i[\tau_j] > 0.$$

Proof. The proof proceeds by showing that the Markov chain's deviation from its limiting distribution, measured by the sums that define the entries of $\tilde{Q}^{\#}$, can be obtained as differences in expected hitting times associated with the process. We begin by expressing $\tilde{Q}^{\#}_{ij}$ in terms of $\tilde{Q}^{\#}_{jj}$, then compute $\tilde{Q}^{\#}_{jj}$. From (3.15), for any i, j,

$$\tilde{Q}^{\#}_{ij} = \lim_{m\to\infty} \sum_{n=0}^{m} ((\Lambda^n)_{ij} - \pi_j).$$

Since $(\Lambda^n)_{ij} = \mathbb{P}_i(Y_n = j) = \mathbb{E}_i[\mathbb{1}(Y_n = j)]$,

$$\sum_{n=0}^{m} ((\Lambda^n)_{ij} - \pi_j) = \mathbb{E}_i[\sum_{n=0}^{m} (\mathbb{1}(Y_n = j) - \pi_j))].$$

Since $\mathbb{1}(Y_n = j) = 0$ for $n < \tau_j$, we can rewrite the above as

$$\begin{aligned} &\mathbb{E}_i\big[\sum_{n=0}^{\tau_j - 1} -\pi_j + \sum_{n=\tau_j}^{m} (\mathbb{1}(Y_n = j) - \pi_j))\big] \\ =&\mathbb{E}_i\big[-\pi_j\tau_j + \sum_{n=\tau_j}^{m} (\mathbb{1}(Y_n = j) - \pi_j))\big]. \end{aligned} \tag{3.17}$$

Then, using the strong Markov property,

$$\mathbb{E}_i[\sum_{n=\tau_j}^{m} (\mathbb{1}(Y_n = j) - \pi_j))] = \sum_{k=0}^{m} \mathbb{P}_i(\tau_j = k)\mathbb{E}_j[\sum_{n=0}^{m-k} (\mathbb{1}(Y_n = j) - \pi_j))].$$

Now separating and using the above relation, (3.17) is equal to

$$\begin{aligned} &-\pi_j\mathbb{E}_i[\tau_j] + \sum_{k=0}^{m} \mathbb{P}_i(\tau_j = k)\mathbb{E}_j\big[\sum_{n=0}^{m-k} (\mathbb{1}(Y_n = j) - \pi_j))\big] \\ =&-\pi_j\mathbb{E}_i[\tau_j] + \sum_{k=0}^{m} \mathbb{P}_i(\tau_j = k) \sum_{n=0}^{m-k} ((\Lambda^n)_{jj} - \pi_j). \end{aligned}$$

Taking the limit as $m \to \infty$,

$$\lim_{m\to\infty} \sum_{k=0}^{m} \mathbb{P}_i(\tau_j = k) \sum_{n=0}^{m-k} ((\Lambda^n)_{jj} - \pi_j) = \sum_{n=0}^{\infty} ((\Lambda^n)_{jj} - \pi_j) = \tilde{Q}^{\#}_{jj}$$

Therefore,

$$\tilde{Q}^{\#}_{ij} = -\pi_j \mathbb{E}_i[\tau_j] + \tilde{Q}^{\#}_{jj} \tag{3.18}$$

Next we obtain $\tilde{Q}^{\#}_{jj}$, which is equal to

$$\lim_{m\to\infty} \sum_{n=0}^{m} ((\Lambda^n)_{jj} - \pi_j).$$

We examine each term separately, beginning with the second one. By stationarity, $\pi_j = \mathbb{P}_\pi(Y_n = j)$ for each n, so

$$\sum_{n=0}^{m} \pi_j = \mathbb{E}_\pi\big[\sum_{n=0}^{m} \mathbb{1}(Y_n = j)\big].$$

Using again the fact that $\mathbb{1}(Y_n = j) = 0$ for $n < \tau_j$, as well as the strong Markov property,

$$\begin{aligned}\mathbb{E}_\pi\big[\sum_{n=0}^{m} \mathbb{1}(Y_n = j)\big] &= \sum_{k=0}^{m} \mathbb{P}_\pi(\tau_j = k)\mathbb{E}_j[\sum_{n=0}^{m-k}(\mathbb{1}(Y_n = j))] \\ &= \sum_{k=0}^{m} \mathbb{P}_\pi(\tau_j = k) \sum_{n=0}^{m-k} (\Lambda^n)_{jj}.\end{aligned}$$

On the other hand,

$$\sum_{n=0}^{m} (\Lambda^n)_{jj} = (\sum_{k=0}^{m} \mathbb{P}_\pi(\tau_j = k) + \mathbb{P}_\pi(\tau_j > m)) \sum_{n=0}^{m} (\Lambda^n)_{jj}.$$

Let $R_m = \mathbb{P}(\tau_j > m)) \sum_{n=0}^{m} (\Lambda^n)_{jj}$ (R is for "remainder"). Combining the two parts above,

$$\begin{aligned}\sum_{n=0}^{m} ((\Lambda^n)_{jj} - \pi_j) &= \sum_{k=0}^{m} \mathbb{P}_\pi(\tau_j = k)\big[\sum_{n=0}^{m} (\Lambda^n)_{jj} - \sum_{n=0}^{m-k} (\Lambda^n)_{jj}\big] + R_m \\ &= \sum_{k=0}^{m} \mathbb{P}_\pi(\tau_j = k)\big[\sum_{n=m-k+1}^{m} (\Lambda^n)_{jj}\big] + R_m.\end{aligned}$$

Using $\lim_{n\to\infty}(\Lambda^n)_{jj} = \pi_j$ and the fact that $\sum_{n=m-k+1}^{m} \cdots$ has k terms, and taking $m \to \infty$, as long as $R_m \to 0$ it will follow that

$$\tilde{Q}^{\#}_{jj} = \sum_{k=0}^{\infty} \mathbb{P}_\pi(\tau_j = k) k \pi_j = \pi_j \mathbb{E}_\pi[\tau_j],$$

and then combining with (3.18) gives (3.16). So, we only need to show that $R_m \to 0$ as $m \to \infty$. Since Λ is assumed to be primitive, there exists $c > 0$ and an integer N such that $\Lambda^N_{ij} \geq c$ for all i, j, which implies $\mathbb{P}_\pi(\tau_j > m) \leq (1-c)^k$ for any $m \geq kN$. Written in terms of m, $\mathbb{P}(\tau_j > m) \leq (1-c)^{m/N-1}$. Since $0 \leq (\Lambda^n)_{jj} \leq 1$, being a probability, $\sum_{n=0}^{m}(\Lambda^n)_{jj} \leq (m+1)$. Together, $0 \leq R_m \leq (m+1)(1-c)^{m/N-1}$ which $\to 0$ as $m \to \infty$. □

3.3.1.2 Stochastic interpretation of perturbation in discrete time

In this section, we will discuss the effect, on the Perron vectors, of perturbing a primitive matrix. We will discuss the general primitive case as well as the special case of primitive stochastic matrices. We will explore how perturbations to matrix A impact the canonical flow of u and v and provide an interpretation of the outcomes.

General Case: Let us say we have a network of individuals and want to model the spread of a disease through this network. To do this, we can represent the network as a matrix A, where A_{ij} is the expected number of type j infections caused by a type i infected individual.

Suppose we want to perturb the network by increasing or decreasing the transmission of infection from k to ℓ. We can represent this perturbation by assuming the derivative A'_{ij} is equal to the following:

$$A'_{ij} = \begin{cases} 1 & (i,j) = (k,\ell) \\ 0 & (i,j) \neq (k,\ell). \end{cases}$$

Next, we want to use equation (3.10) to compute the perturbation of v_i. Equation (3.10) gives us:

$$v'_i = \frac{Q^{\#}_{ik} A'_{k\ell} v_\ell}{r}.$$

It turns out to be convenient to examine the relative change in v_i, so we write

$$v'_i/v_i = Q^{\#}_{ik}(v_\ell/v_i)/r.$$

Now to connect this with the expected hitting time, we can rewrite the equation as follows:

$$\begin{aligned} v_i'/v_i &= Q^{\#}_{ik}(v_k/v_i)(v_\ell/v_k)/r \\ &= (I-\Lambda)^{\#}_{ik}\,\frac{v_\ell}{rv_k} \text{ where } \Lambda_{ik} = (A_{ik}/r)v_k/v_i \\ &= \pi_k(\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k])\frac{v_\ell}{rv_k} \\ &= (\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k])\frac{u_k v_\ell}{r}, \end{aligned}$$

using the fact that $\pi_k = v_k u_k$ on the last line. This expression tells us if $\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k]$ is negative, then v_i'/v_i is negative, i.e., the potential of type i is decreasing. $\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k]$ represents the expected time it takes the Markov chain to reach k starting with the initial distribution π (which represents the long-term behaviour of the disease spread in the network), minus the expected time it takes for the Markov chain to reach k, starting from i (which represents the short-term behaviour of the disease spread starting from i). If $\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k]$ is negative, this means that the Markov chain can reach k faster when it is initialized in the stationary distribution than it can if it is initialized at i. Another way to say this is that it takes 'longer than average' to reach k from i.

Now we will see how we can interpret the perturbation of u in terms of expected hitting times using (3.19). In order to use the same approach as above we need to use the time-reversal of the previous Markov chain. To start with, we need to derive formulas analogous to those for $(I-\Lambda)^{\#}_{ii}$ and $(I-\Lambda)^{\#}_{ij}$. Time-reversing amounts to stochasticizing from the other direction, as follows: define $C_{ij} = (u_i/u_j)(A_{ij}/r)$. Then

$$(I-C)^{\#}_{ij} = \sum_{n=0}^{\infty}(C^n_{ij} - \pi_i)$$

with π as in the other case. In addition,

$$\sum_i C_{ij} = \sum_i u_i A_{ij}(u_j/r) = 1.$$

So C is column-stochastic, i.e., $C^\top$ is a stochastic matrix. So, we can get formulas in terms of a time-reversed process obtained after biasing on u. It is not hard to show that π is the stationary distribution of $C^\top$.

Claim: Define π by $\pi_j = u_j v_j$. Then π is the stationary distribution of $C^\top$.

Proof. Since $C_{ij} = (u_i/u_j)(A_{ij}/r)$ is a column stochastic matrix, we have $\sum_i C_{ij} = 1$. Therefore

$$\sum_i \pi_i C^\top_{ij} = \sum_i (v_i u_i)(u_j/u_i)(A_{ji}/r) = u_j \sum_i \frac{A_{ji} v_i}{r} = v_j u_j = \pi_j.$$

which proves our claim. □

From Theorem 3.19 we have the following formulas:

$$(I - C)^\#_{ij} = \pi_i (\tilde{\mathbb{E}}_\pi[\tau_i] - \tilde{\mathbb{E}}_j[\tau_i])$$

where the notation $\tilde{\mathbb{E}}[\cdots]$ denotes expectation with respect to the time-reversed Markov chain. The difference formula

$$(I - C)^\#_{ii} - (I - C)^\#_{ij} = \pi_i \tilde{\mathbb{E}}_j[\tau_i].$$

The relationship with Q is that $(u_i/u_j) Q^\#_{ij} = (I - C)^\#_{ij}$. First we give the formula for u' with the given choice of A':

$$u'_i = u_k \left(\frac{A'_{k\ell}}{r} \right) Q^\#_{\ell i}.$$

Now expanding and then dividing by u_i we get

$$\begin{aligned} u'_i/u_i &= (1/r)(u_k/u_\ell)(u_\ell/u_i) Q^\#_{\ell i} \\ &= (I - C)^\#_{\ell i} \frac{u_k}{r u_\ell} \text{ where } C_{\ell i} = (u_\ell/u_i)(A_{\ell i}/r) \\ &= \pi_\ell (\tilde{\mathbb{E}}_\pi[\tau_\ell] - \tilde{\mathbb{E}}_i[\tau_\ell]) \frac{u_k}{r u_\ell} \\ &= (\tilde{\mathbb{E}}_\pi[\tau_\ell] - \tilde{\mathbb{E}}_i[\tau_\ell]) \frac{u_k v_\ell}{r}, \end{aligned}$$

using the fact that $\pi_\ell = u_\ell v_\ell$ on the last line. This equation shows that the prevalence of the disease at node i is influenced by the expected hitting times of the Markov chain from initial distribution π and node ℓ, as well as the infection potential between nodes k and ℓ. Specifically, the value of $(\tilde{\mathbb{E}}_\pi[\tau_\ell] - \tilde{\mathbb{E}}_i[\tau_\ell])$ represents the expected time it takes for the Markov chain to reach node ℓ from π minus the expected time it takes to reach k starting from i (which relates to the short-term behaviour of the disease spread in

the network).

Special case: To study the case where A is both primitive and stochastic, we want to perturb in a way that preserves that property, i.e., that the row sums remain 1. To do so, this, consider the simple perturbation $A'_{kl} = 1$ and $A'_{kj} = -1$. Since in the stochastic case $r = 1$, using (3.11) we get

$$u' = uA'Q^{\#}.$$

Rewriting somewhat,

$$\begin{aligned} u'_i &= u_k A'_{k\ell} Q^{\#}_{\ell i} + u_k A'_{kj} Q^{\#}_{ji} \\ &= u_k Q^{\#}_{\ell i} - u_k Q^{\#}_{ji} \\ &= u_k (Q^{\#}_{\ell i} - Q^{\#}_{ji}) \\ &= u_k \pi_i \left((\mathbb{E}_\pi[\tau_i] - \mathbb{E}_\ell[\tau_i]) - (\mathbb{E}_\pi[\tau_i] - \mathbb{E}_j[\tau_i]) \right) \\ &= u_k u_i (\mathbb{E}_j[\tau_i] - \mathbb{E}_\ell[\tau_i]), \quad \text{since } \pi_i = u_i. \end{aligned}$$

This expression gives the change in the proportion of time spent at node i due to a small perturbation in the transition matrix A. The interpretation is similar to the general case above.

3.3.2 Continuous time perturbation analysis

This section will cover the $Q^{\#}$ formula and the interpretation of the perturbation effect in the context of hitting time for a continuous-time Markov chain. Both the general case of irreducible Metzler matrix and the special case of irreducible generator matrix will be discussed.

3.3.2.1 $Q^{\#}$ formula in continuous time

Let B be an irreducible Metzler matrix. Recall the interpretation: B_{ij} represents the rate of production of type j by type i for $j \neq i$ and $B_{ii} = b_i - d_i$ where b_i, d_i are respectively the rate of production of type i by type i, and the death rate of type i. For any matrix M let $\alpha(M)$ denote the spectral abscissa of M, i.e., $\alpha(M) = \max\{\text{Real}(\lambda) \colon \lambda \in \sigma(M)\}$ (the largest real part of any eigenvalue of M). Then $B - \alpha(B)I$ has spectral abscissa 0. So in this setting we take $Q = B - \alpha(B)I$. Let u, v denote left and right Perron vectors of e^B, as given by Corollary 3.12, then $e^B v = e^{\alpha(B)I} v$ i.e. $(B - \alpha(B)I)v = 0$ or $Qv = 0$. Differentiating this equation gives

$$Q'v + Qv' = 0$$

so rearranging,

$$Qv' = -Q'v. \tag{3.19}$$

Since $Qv = 0$, Q is not invertible. As before, we want a matrix $Q^{\#}$ that satisfies $QQ^{\#} = Q^{\#}Q = I - P$, where $P = vu/uv$. In this setting, it will be given by an integral rather than a sum. We'll begin with the desired integral, then explain why it is finite and has the desired property:

$$Q^{\#} := -\int_0^\infty (\exp(Qt) - P)dt = -\int_0^\infty (\exp(Qt) - \frac{vu}{uv})dt. \tag{3.20}$$

Lemma 3.20. *Let B, Q be as above. The integral in* (3.20) *is convergent and $Q^{\#}$ satisfies $QQ^{\#} = Q^{\#}Q = I - P$.*

Proof. The reasoning is similar to the proof of Lemma 3.14. Since B is irreducible Metzler, Q is as well (it is only different from B on the diagonal). Since Q and e^{Qt} have the same eigenspaces, and since e^{Qt} is primitive, Q has Jordan form

$$Q = S \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & K \end{array} \right) S^{-1},$$

for some invertible S and some K with $\alpha(K) < 0$, and e^{Qt} has Jordan form

$$e^{Qt} = S \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & e^{Kt} \end{array} \right) S^{-1}.$$

Then,

$$P = S \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & 0 \end{array} \right) S^{-1}$$

so

$$e^{Qt} - P = S \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & e^{Kt} \end{array} \right) S^{-1}.$$

Since $\alpha(K) < 0$, $\int_0^\infty e^{Kt}dt$, the resolvent integral generated by K, is convergent and satisfies $\int_0^\infty e^{Kt}dt = -K^{-1}$. It follows that $\int_0^\infty (e^{Qt} - P)dt$ is convergent and that

$$Q^{\#} = -S \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & -K^{-1} \end{array} \right) S^{-1} = S \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & K^{-1} \end{array} \right) S^{-1}.$$

From this formula it is clear that

$$QQ^{\#} = Q^{\#}Q = S \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & I \end{array} \right) S^{-1}$$

and comparing to the formula for P, this is equal to $I - P$. □

Making the same assumption $u'v = uv' = 0$ as before, multiplying (3.19) on the left by $Q^{\#}$ and using $Q^{\#}Q = I - P$ and $Pv' = (v/uv)uv' = 0$,

$$v' = -Q^{\#}Q'v = -Q^{\#}B'v, \tag{3.21}$$

using $Q' = B' - \alpha(B)'$ and $Q^{\#}v = 0$ (note that $\alpha(B)'$ is a scalar). Similarly,

$$u' = -uQ'Q^{\#} = -uB'Q^{\#}. \tag{3.22}$$

The formula for $Q^{\#}$ can be interpreted in terms of a Markov chain in a similar way as in discrete time in Theorem 3.19 but in order to do this we need to make sure that in addition to spectral abscissa of Q is 0, B has zero row sums, that is, $B\mathbf{1} = 0$ where $\mathbf{1} = (1, \dots, 1)^{\top}$. Equivalently, B generates a continuous-time Markov chain, i.e., for $t \in \mathbb{R}_+$ e^{Bt} is a stochastic matrix. If B does not satisfy this condition then we need to do the analogue of stochasticization, just like the discrete time case. Let us define $W_{ij} = Q_{ij}v_j/v_i = (B_{ij} - \alpha(B))v_j/v_i$ so that W has zero row sums. Then, if u, v are chosen so that $uv = 1$ then π defined by $\pi_j = u_j v_j$ is the stationary distribution of the Markov chain generated by W (it satisfies $\pi W = 0$).

Using (3.20), W has the formula

$$W^{\#}_{ij} = -\int_0^{\infty} (e^{Ws}_{ij} - \pi_j) ds. \tag{3.23}$$

If W generates a continuous-time Markov chain (Y_t) then $e^{Wt}_{ij} = \mathbb{P}_i(Y_t = j)$, since e^{Wt} solves the Kolmogorov equations for the transition probability of the Markov chain. Theorem 3.19 can be restated in terms of continuous time as follows.

Theorem 3.21. *Let W be an irreducible generator matrix with stationary distribution π. Then $-W^{\#}$ satisfies the same formulas that $\tilde{Q}^{\#}$ satisfies in Theorem 3.19, where $\mathbb{P}, \mathbb{E}$ are with respect to the continuous-time Markov chain with generator W.*

Proof. The reasoning is the same as the proof of Theorem 3.19, only Λ^n is replaced with e^{Ws} and the sums are replaced with integrals, so we will just sketch out the first few steps, to make the analogy clear. Let (Y_t) denote the continuous-time Markov chain with generator matrix W. Then,

$$\int_0^t (e_{ij}^{Ws} - \pi_j)ds = \mathbb{E}_i\big[\int_0^t (\mathbb{1}(Y_s = j) - \pi_j)ds\big].$$

With $\tau_j = \inf\{t \colon Y_t = j\}$, since $\mathbb{1}(Y_s = j) = 0$ for $s < \tau_j$ this is equal to

$$\mathbb{E}_i\big[-\pi_j\tau_j + \int_{\tau_j}^t (\mathbb{1}(Y_s = j) - \pi_j)ds\big].$$

As before, separate and use the strong Markov property. Following the same steps as in the proof of Theorem 3.19 and noting the additional minus sign that appears in the formula for the group inverse in continuous time, we get

$$-W_{ij}^{\#} = -\pi_j\mathbb{E}_i[\tau_j] - W_{jj}^{\#}.$$

The proof of the other formula similarly follows the exact same steps as the proof of Theorem 3.19; note that the bound on τ_j can be deduced in the same way using powers of the primitive matrix e^W. □

3.3.2.2 Stochastic interpretation of perturbation in continuous time

In this section, which is analogous to Section 3.3.1.2, we discuss the effect of perturbing an irreducible Metzler matrix on its Perron vectors. We discuss both the general case and the case of a generator matrix.

General Case: Suppose we want to perturb the matrix B by changing the rate of spread from k to ℓ. This can be understood by taking the following for the derivative B':

$$B'_{ij} = \begin{cases} 1 & (i,j) = (k,\ell) \\ 0 & (i,j) \neq (k,\ell). \end{cases}$$

Letting $Q = B - \alpha(B)I$ as before and using (3.21) with the given choice of B' we get

$$v_i'/v_i = -Q_{ik}^{\#}(v_\ell/v_i).$$

Now to connect this with the expected hitting time, we can rewrite the equation as follows, using $W_{ij} = Q_{ij}v_j/v_i$ as before:

$$\begin{aligned} v_i'/v_i &= -Q^{\#}_{ik}(v_k/v_i)(v_\ell/v_k) \\ &= -W^{\#}_{ik}\,\frac{v_\ell}{v_k} \\ &= \pi_k(\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k])\frac{v_\ell}{v_k} \\ &= (\mathbb{E}_\pi[\tau_k] - \mathbb{E}_i[\tau_k])u_k v_\ell, \text{ since } \pi_k = u_k v_k. \end{aligned}$$

Now we will see how we can interpret the perturbation of u in terms of expected hitting times using (3.21). Again we need to use the time reversal method to understand the effect of the perturbation on the left Perron vector. To start with, we need to derive formulas analogous to those for $-W^{\#}_{ii}$ and $-W^{\#}_{ij}$. To do so, we need to stochasticize from the other direction. In other words, define instead $D_{ij} = (u_i/u_j)Q_{ij}$. Then D has zero column sums, i.e., $D^\top$ is a generator matrix. So, we can get formulas in terms of a time-reversed process obtained after biasing on u. The proof that π is the stationary distribution of the Markov chain with generator $D^\top$ is analogous to the discrete-time case.

From Theorem 3.21 we have the following formulas:

$$-D^{\#}_{ij} = \pi_i\tilde{\mathbb{E}}_\pi[\tau_i] - \pi_i\tilde{\mathbb{E}}_j[\tau_i].$$

The difference formula is

$$-D^{\#}_{ii} + D^{\#}_{ij} = \pi_i\tilde{\mathbb{E}}_j[\tau_i].$$

The relationship with Q is that $(u_i/u_j)Q^{\#}_{ij} = D^{\#}_{ij}$. So from the perturbation formula $u' = -uB'Q^{\#}$ as before we get

$$u_i' = -u_k B'_{kl} Q^{\#}_{\ell i},$$

then expanding and then dividing by u_i we get

$$\begin{aligned} u_i'/u_i &= -(u_k/u_\ell)(u_\ell/u_i)Q^{\#}_{\ell i} \\ &= -D^{\#}_{\ell i}\,\frac{u_k}{u_\ell} \\ &= \pi_\ell(\tilde{\mathbb{E}}_\pi[\tau_\ell] - \tilde{\mathbb{E}}_i[\tau_\ell])\frac{u_k}{u_\ell} \\ &= (\tilde{\mathbb{E}}_\pi[\tau_\ell] - \tilde{\mathbb{E}}_i[\tau_\ell])u_k v_\ell, \text{ since } \pi_\ell = u_\ell v_\ell. \end{aligned}$$

Special Case: In this case we will just compute u_i' since this is all we need for the application that follows; v_i' is analogous. Suppose W is a generator matrix with right eigenvector $v = \mathbf{1}=(1,1,1,\cdots 1)$ and stationary distribution $u = \pi$ satisfying $\pi W = 0$ and $\pi v = 1$, and for fixed $k \neq \ell$, consider the simple perturbation

$$W'_{k\ell} = 1 \text{ and } W'_{kk} = -1.$$

Using (3.22) with the given choice of W',

$$\begin{aligned} u_i' &= -u_k W'_{k\ell} W^{\#}_{\ell i} - u_k W'_{kk} W^{\#}_{ki} \\ &= -u_k W^{\#}_{\ell i} + u_k W^{\#}_{ki} \\ &= u_k(-W^{\#}_{\ell i} + W^{\#}_{ki}) \\ &= u_k \pi_i \left((\mathbb{E}_\pi[\tau_i] - \mathbb{E}_\ell[\tau_i]) - (\mathbb{E}_\pi[\tau_i] - \mathbb{E}_k[\tau_i]) \right) \\ &= u_k u_i (\mathbb{E}_k[\tau_i] - \mathbb{E}_\ell[\tau_i]), \quad \text{since} \quad \pi_i = u_i. \end{aligned} \tag{3.24}$$

3.3.3 Application: results of [38]

Our work can be used to recover some results of [38] with less effort, so we begin by introducing their model. In an earlier article [21] the authors take a fast-movement limit of a multi-patch modified SIR model of Cholera spread and in the limit obtain an expression for the basic reproduction number (explained in the Introduction) of the model that has the following form:

$$R_0 = \sum_i R_0^{(i)} u_i$$

where i indexes patches, $R_0^{(i)}$ is the (local) basic reproduction number at patch i (some known values), and (u_i) is the stationary distribution of the continuous-time Markov chain with a generator matrix that we will denote by W. With respect to their notation, for $i \neq j$, $W_{ij} = m_{ji}$ where m_{ji} is the movement rate of the pathogen from patch i to j, and W_{ii} is chosen to make the rows sum to 0, i.e., $W_{ii} = -\sum_{j \neq i} m_{ji}$. In their notation, u is the stationary distribution of W, i.e., it satisfies $uW = 0$. The goal of their work is to understand how changes in the movement rates m_{ji} affects the entries in u_i, which in turn affects R_0. The authors of [38] obtained their formula for $\partial u_i / \partial m_{\ell k}$ for arbitrary i, k, ℓ using both combinatorial methods and algebraic methods (see their equations (3.7) and (4.3) respectively); in terms of our notation this formula corresponds to the first line of (3.24) (the given choice of W' means that u_i' corresponds to $\partial u_i / \partial m_{\ell k}$). The authors

use this formula to determine the sign of u_i' (Theorem 5.1 of [38]) for specific i, k, ℓ in the context of a star network. We will obtain this formula more quickly using the last line of (3.24). We describe the context: a star network has nodes $1, \ldots, n$ and edges $\{1, j\}$ for all $j > 1$; node 1 is called the hub and nodes $2, \ldots, n$, the leaves. The authors used a combinatorial method (counting spanning rooted trees) to prove Theorem 5.1 in a star network for hub 1 and leaves i, k and ℓ. In our notation, the result is as follows: with the perturbation $W'_{k\ell} = 1,\ W'_{kk} = -1$,

$$\operatorname{sgn}(u_i') = \operatorname{sgn}(W_{\ell 1} - W_{k1}),$$

in other words that the proportion of time spent by pathogens at i increases if and only if the movement rate from ℓ to 1 is larger than from k to 1. We can recover this from our formula as follows. Note that to reach any leaf from any other leaf, it is necessary to pass through the hub node 1, as shown in Figure 3.2.

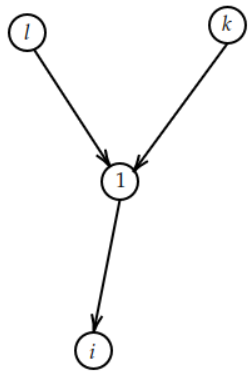


Figure 3.2: A depiction of the path taken from leaf nodes ℓ and k to leaf node i, through the hub node 1 in a star network.

We then have

$$u_i' = u_k u_i (\mathbb{E}_k[\tau_i] - \mathbb{E}_\ell[\tau_i]).$$

Since u_k and u_i are positive, $\operatorname{sgn}(u_i') = \operatorname{sgn}(\mathbb{E}_k[\tau_i] - \mathbb{E}_\ell[\tau_i])$. Moreover, for any leaf node k, $\mathbb{E}_k[\tau_1] = 1/W_{k1}$, since the time to travel from k to 1 is

exponentially distributed with rate W_{k1}. Now

$$\begin{aligned}\mathbb{E}_k[\tau_i] - \mathbb{E}_\ell[\tau_i] &= \mathbb{E}_k[\tau_1] + \mathbb{E}_1[\tau_i] - \mathbb{E}_\ell[\tau_1] - \mathbb{E}_1[\tau_i] \\ &= \mathbb{E}_k[\tau_1] - \mathbb{E}_\ell[\tau_1] \\ &= \frac{1}{W_{k1}} - \frac{1}{W_{\ell 1}} \\ &= \frac{W_{\ell 1} - W_{k1}}{W_{\ell 1} W_{k1}}.\end{aligned}$$

Since $W_{\ell 1}, W_{k1}$ are positive, $\operatorname{sgn}(u_i') = \operatorname{sgn}(W_{\ell 1} - W_{k1})$ as desired.

Chapter 4

Slow-fast decomposition

The main goal of this chapter is to explore the slow-fast decomposition of a structured SIR model in a controlled environment, in the sense described below, and to demonstrate its connection with the canonical flow equations introduced in Chapter 3.

Our approach begins with a motivating example that demonstrates how the slow-fast decomposition works in the context of a logistic equation. From there, we shift our focus to a structured SIR model in which the S, R variables are slow and the I variables are fast. By assuming that I is small relative to S (small-I), and that on average there is no exponential growth or decay of I (criticality), we are able to establish a connection between the structured SIR model and the canonical flow equations discussed in Chapter 3. Finally, we perform a numerical simulation of an age-structured SIR model with two age classes to observe how the dynamics of disease behave in a slow-fast decomposition with a controlled environment.

Researchers in various scientific fields, such as neuroscience, physics, and biology, use a mathematical method called slow-fast decomposition to simplify complex systems into two parts: a fast component that changes quickly and a slow component that changes slowly. This method helps researchers identify the most dominant behaviours in a system by dividing its dynamics into fast and slow time scales. By isolating the interactions that create the overall behaviour of a complex system, researchers can gain a deeper understanding of it. Analyzing these interactions also allows them to predict the system's future behaviour.

Consider the logistic equation with a slowly decreasing parameter:

$$\begin{aligned} d\lambda/dt &= -\epsilon, \\ dx/dt &= \lambda x - x^2. \end{aligned}$$

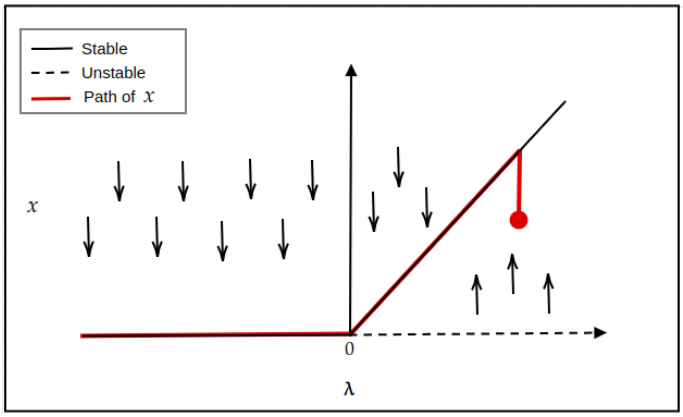


Figure 4.1: The x variablde changes quickly to approach the slow manifold and changes slowly along the slow manifold as λ is varied.

Taking $t = \epsilon\tau$ the equations become

$$d\lambda/d\tau = -1,$$
$$\epsilon dx/d\tau = \lambda x - x^2.$$

When ϵ is small, the dynamics of x are much faster than the dynamics of λ. This means that x quickly approaches the slow manifold, which is the graph of the piecewise function

$$f(\lambda) = \begin{cases} 0 & \text{if } \lambda \leq 0, \\ \lambda & \text{if } \lambda > 0 \end{cases}$$

that gives the asymptotically stable (from any positive initial condition) equilibrium of the dx/dt equation, as a function of λ.

Once x is on the slow manifold, it changes much more slowly than λ. This means that λ changes relatively slowly while x remains close to the slow manifold. This gives rise to the slow-fast behaviour of the system, where x changes quickly to approach the slow manifold and then changes slowly along the slow manifold as λ changes slowly.

In the upcoming section, we will see the slow-fast analysis of the structured SIR model. We will demonstrate how the canonical flow equation (3.11), as introduced in Chapter 3, emerges naturally when certain constraints are imposed. Additionally, we will conduct a numerical analysis of

the controlled slow-fast structured SIR model, offering insights into disease dynamics.

4.1 Slow-fast decomposition of a structured SIR model

We recall the structured SIR model discussed in the Introduction:

$$\begin{aligned} S_j' &= -\sum_{i=1}^{n} I_i \beta_{ij} S_j, \\ I_j' &= \sum_{i=1}^{n} I_i \beta_{ij} S_j - I_j \gamma_j, \\ R_j' &= I_j \gamma_j, \end{aligned} \tag{4.1}$$

where $\beta_{ij} \geq 0$ is the transmission rate from type i to type j and $\gamma_j > 0$ is the recovery rate of type j. We think of $(S_j), (I_j), (R_j)$ as row vectors. Using the $\circ$ symbol to denote element-wise multiplication, this can be written in matrix form as

$$\begin{aligned} S' &= -I\beta \circ S, \\ I' &= I\beta \circ S - I \circ \gamma, \\ R' &= I \circ \gamma. \end{aligned} \tag{4.2}$$

We make the following two assumptions, which we discuss briefly.

Criticality. When an outbreak occurs, various control measures or restrictions (such as quarantine, vaccination, closure etc.) can be introduced to prevent the spread of infection, and these can be modelled by allowing the transmission rates β_{ij} to vary over time, i.e., letting $\beta_{ij} = \beta_{ij}(t)$. Typically, these restrictions are intended to keep infections below a certain threshold and are lifted or reduced once this threshold is met, causing infections to increase again. This might be expressed mathematically as follows: there exist $c, C > 0$ such that for all j, t,

$$c \sum_j I_j(0) \leq \sum_j I_j(t) \leq C \sum_j I_j(0). \tag{4.3}$$

In other words, $\sum_j I_j(t)$ does not experience any exponential growth or

decay over time. The I equations are

$$\begin{aligned} I' &= I\beta \circ S - I \circ \gamma \\ &= I(\beta \circ S - \mathrm{diag}(\gamma)) \\ &= I\psi, \end{aligned} \tag{4.4}$$

where $\mathrm{diag}(\gamma)$ is the diagonal matrix with entries $D_{ii} = \gamma_i$ and $\psi = \beta \circ S - \mathrm{diag}(\gamma)$. Since $I' = I\psi$, the above comment suggests that $\alpha(\psi) = \max\{\mathrm{Real}(\lambda) : \lambda \in \sigma(\psi)\}$ remains close to 0 in an average sense. In any case, we can make it 0 with a change of variables: letting $\tilde{\psi} = \psi - \alpha(\psi)$ and $\tilde{I}(t) = \exp(-\int_0^t \alpha(\psi(s))ds)I(t)$ we get, $\alpha(\tilde{\psi}) = 0$ and we can check that $d\tilde{I}/dt = \tilde{I}\tilde{\psi}$, as follows:

$$\begin{aligned} \frac{d\tilde{I}}{dt} &= \frac{d}{dt}\left(\exp\left(-\int_0^t \alpha(\psi(s))ds\right) I(t)\right) \\ &= \exp\left(-\int_0^t \alpha(\psi(s))ds\right)\frac{dI}{dt} - \alpha(\psi(t))\exp\left(-\int_0^t \alpha(\psi(s))ds\right) I(t) \\ &= \tilde{I}\left(\psi(t) - \alpha(\psi(t))\right) \\ &= \tilde{I}\tilde{\psi}. \end{aligned}$$

So, in what follows, we will assume that $\alpha(\psi) = 0$.

Small-I approximation. Even during relatively large outbreaks, the fraction of the population that is infected tends to be fairly low. For example, during the COVID-19 pandemic, which began in late 2019 and has affected millions of people worldwide, the fraction of the Canadian population that has been infected when the number of infectious is at largest is around 0.7% of the total population; see the reports of [9].

If we assume that $I(0) = \epsilon\, u_0$ where $u_0 \neq 0$ and make the change of variables $\mathcal{I} = I/\epsilon$, where ϵ is a small, positive parameter, then $\mathcal{I}_0 = u_0$ and the system becomes

$$\begin{aligned} S' &= -\epsilon\mathcal{I}\beta \circ S, \\ \mathcal{I}' &= \mathcal{I}\psi, \\ R' &= \epsilon\mathcal{I} \circ \gamma. \end{aligned}$$

Let $\tau = \epsilon t$. Then, τ represents the slow time scale. We use the following notation for state variables on the slow time scale to emphasize the dependence

on ϵ:

$$\begin{aligned}\mathcal{S}^\epsilon(\tau) &= S^\epsilon(\tau/\epsilon),\\ \mathcal{I}^\epsilon(\tau) &= I^\epsilon(\tau/\epsilon)/\epsilon,\\ \mathcal{Q}^\epsilon(\tau) &= \psi^\epsilon(\tau/\epsilon),\\ \mathcal{R}^\epsilon(\tau) &= R^\epsilon(\tau/\epsilon),\\ \mathcal{B}^\epsilon(\tau) &= \beta^\epsilon(\tau/\epsilon).\end{aligned}$$

The system can be rewritten as

$$\begin{aligned}\mathcal{S}^{\epsilon\prime} &= -\mathcal{I}^\epsilon \mathcal{B}^\epsilon \circ \mathcal{S}^\epsilon,\\ \mathcal{I}^{\epsilon\prime} &= \frac{1}{\epsilon}\mathcal{I}^\epsilon \mathcal{Q}^\epsilon,\\ \mathcal{R}^{\epsilon\prime} &= \mathcal{I}^\epsilon \circ \gamma\end{aligned} \tag{4.5}$$

with initial conditions specified by $\mathcal{I}^\epsilon(0) = u_0$ and $\mathcal{S}^\epsilon(0) = \mathcal{S}_0$. Setting the total population equal to 1 determines $\mathcal{R}^\epsilon(0)$.

Let us first examine the equation for the fast variable $\mathcal{I}^\epsilon$. To understand it, it will be helpful to let $u^\epsilon(\tau), v^\epsilon(\tau)$ denote the solutions to the canonical flow equations with τ in place of t, $\mathcal{Q}^\epsilon(\tau)$ in place of $Q(t)$, $u^\epsilon(0) = u_0$ and any choice of $v^\epsilon(0)$ that satisfies $\mathcal{Q}^\epsilon(0)v^\epsilon(0) = 0$. If $\mathcal{Q}^\epsilon$ does not change too quickly, then since $\alpha(\mathcal{Q}^\epsilon) = 0$ and the 0-eigenspace is one-dimensional, because of the $1/\epsilon$ multiplier $\mathcal{I}^\epsilon$ is pushed quickly towards equilibrium, i.e., $\mathcal{I}^\epsilon(\tau)\mathcal{Q}^\epsilon(\tau) \approx 0$ for all τ. Also, since $\mathcal{Q}^\epsilon(\tau)v^\epsilon(\tau) = 0$,

$$\mathcal{I}^{\epsilon'}(\tau)v^\epsilon(\tau) = \frac{1}{\epsilon}\mathcal{I}^\epsilon(\tau)\mathcal{Q}^\epsilon(\tau)v^\epsilon(\tau) = 0 \quad \forall\tau.$$

Since $\mathcal{I}^{\epsilon'}(\tau)v^\epsilon(\tau) = 0$ and $\mathcal{I}^\epsilon(\tau)\mathcal{Q}^\epsilon(\tau) \approx 0$, which are the two parts of the characterization of the canonical flow discussed in subsection 3.2.2, it is reasonable to suppose that $\mathcal{I}^\epsilon \approx u^\epsilon$. If, moreover, $\mathcal{Q}^\epsilon$ converges as $\epsilon \to 0$ to a function $\mathcal{Q}$, then one might suppose that $\mathcal{I}^\epsilon$ converges to the left part of the canonical flow corresponding to $\mathcal{Q}$ with initial value u_0.

Having understood the fast variable, we turn to the full set of equations (4.5) on the slow time scale and try to find a suitable limit as $\epsilon \to 0$. Borrowing the left part (3.13) of the canonical flow equations and noting that the $\mathcal{S}^{\epsilon\prime}$ and $\mathcal{R}^{\epsilon\prime}$ equations do not depend explicitly on ϵ, a suitable set

of equations representing the limit of (4.5) as $\epsilon \to 0$ is given by

$$\begin{aligned} \mathcal{S}' &= -\mathcal{I}\mathcal{B} \circ \mathcal{S}, \\ \mathcal{I}' &= -\mathcal{I}\mathcal{Q}'\mathcal{Q}^{\#}, \\ \mathcal{R}' &= \mathcal{I} \circ \gamma. \end{aligned} \tag{4.6}$$

In words, $\mathcal{S}$ and $\mathcal{R}$ follow the same equation (with $\mathcal{I}$ in place of I) as on the original time scale, while $\mathcal{I}$ follows the left part of the canonical flow (3.11) associated with $\mathcal{Q}$. This shows the connection between the slow-fast structured SIR model and the perturbation formula we get from chapter 3. Notice that if $\mathcal{B}^\epsilon$ and $\mathcal{S}^\epsilon$ converge to matrix- and vector-valued functions $\mathcal{B}$ and $\mathcal{S}$ as $\epsilon \to 0$ then $\mathcal{Q}^\epsilon \to \mathcal{Q} := \mathcal{B} \circ \mathcal{S} - \operatorname{diag}(\gamma)$. In the context of (4.5), $\mathcal{B}^\epsilon$ is an external input while $\mathcal{S}^\epsilon$ is a (vector-valued) state variable. The convergence of $\mathcal{B}$ means that the effective transmission rate converges on the slow time scale. This motivates the following conjecture.

Conjecture 4.1. For $\epsilon > 0$ let $(\mathcal{S}^\epsilon, \mathcal{I}^\epsilon, \mathcal{R}^\epsilon)$ solve (4.5) with initial conditions

$$(\mathcal{S}^\epsilon(0), \mathcal{I}^\epsilon(0), \mathcal{R}^\epsilon(0)) = (\mathcal{S}_0, u_0, \mathcal{R}_0),$$

where $\mathcal{B}^\epsilon$ is a continuous matrix-valued function of τ and $\alpha(\mathcal{Q}^\epsilon(\tau)) = 0$ for all τ. If there is a continuous function $\mathcal{B}$ such that $\sup_{\tau \le T} |\mathcal{B}^\epsilon(\tau) - \mathcal{B}(\tau)| \to 0$ for each $T > 0$ as $\epsilon \to 0$, then letting $(\mathcal{S}, \mathcal{I}, \mathcal{R})$ denote the solutions to (4.6) with the same initial conditions, for each $T > 0$,

$$\begin{aligned} &\sup_{\tau \le T} |\mathcal{S}^\epsilon(\tau) - \mathcal{S}(\tau)| \to 0, \\ &\sup_{\tau \le T} |\mathcal{I}^\epsilon(\tau) - \mathcal{I}(\tau)| \to 0 \quad \text{and} \\ &\sup_{\tau \le T} |\mathcal{R}^\epsilon(\tau) - \mathcal{R}(\tau)| \to 0 \end{aligned}$$

as $\epsilon \to 0$.

4.1.1 Numerical algorithm

In this subsection, we present a convenient numerical algorithm for simulating the system (4.6), and show a couple of simulation plots to get an idea of how the system is working.

First, we will start with the numerical algorithm. Note that the term $\mathcal{Q}'$ appearing on the right-hand side of the $\mathcal{I}(\tau)$ equation can lead to instability

if not handled carefully. So we want to compute $\mathcal{I}'$ in a different way than just applying a standard algorithm to the given vector field. We propose to use the following iteration for $\mathcal{I}_n$:

$$\mathcal{I}_{n+1} = \mathcal{I}_n P_{n+1}, \tag{4.7}$$

where P_n is the usual projection matrix i.e. $P_n = v_n u_n / u_n v_n$, where $u_n \mathcal{Q}_n = \mathcal{Q}_n v_n = 0$. The question arises: why should this approach be effective? It can be demonstrated that if $(\mathcal{I}_n)$ is chosen in this way then it satisfies the discrete analogue of the two-part characterization of the canonical flow driven by $\mathcal{Q}_n$ as described by Theorem 3.18, namely,

$$\mathcal{I}_n \mathcal{Q}_n = 0 \text{ and } (\mathcal{I}_{n+1} - \mathcal{I}_n) v_{n+1} = 0.$$

To verify the first property, using the definition (4.7),

$$\mathcal{I}_{n+1} = \mathcal{I}_n P_{n+1} = \mathcal{I}_n v_{n+1} u_{n+1} / u_{n+1} v_{n+1} = (\mathcal{I}_n v_{n+1} / u_{n+1} v_{n+1}) u_{n+1}.$$

Multiplying both sides on the right by $\mathcal{Q}_{n+1}$ and noting that $u_{n+1}\mathcal{Q}_{n+1} = 0$ obtains the conclusion. For the second property,

$$\begin{aligned} \mathcal{I}_{n+1} - \mathcal{I}_n &= \mathcal{I}_n (P_{n+1} - I) \\ (\mathcal{I}_{n+1} - \mathcal{I}_n) v_{n+1} &= \mathcal{I}_n (P_{n+1} - I) v_{n+1}. \end{aligned}$$

Since $P_n v_n = v_n$ for all n, the right side of the above equation is 0. Therefore $(\mathcal{I}_{n+1} - \mathcal{I}_n) v_{n+1} = 0$ as desired. To compute $(\mathcal{S}_n), (\mathcal{R}_n)$ we simply use the first-order Euler method, as we do not expect much sensitivity. Altogether our numerical algorithm for approximating solutions of (4.6) by $(\mathcal{S}_n, \mathcal{I}_n) \approx (\mathcal{S}(nh), \mathcal{I}(nh))$ is

$$\begin{aligned} \mathcal{S}_{n+1} &= \mathcal{S}_n + h \cdot (-\mathcal{I}_n \mathcal{B}_n \circ \mathcal{S}_n) \\ \mathcal{I}_{n+1} &= \mathcal{I}_n P_{n+1} \\ \mathcal{R}_{n+1} &= \mathcal{R}_n + h \cdot (\mathcal{I}_n \circ \gamma). \end{aligned} \tag{4.8}$$

4.1.2 Example and numerical simulation

Now, let us consider a numerical example to illustrate our model (4.8). Suppose we have a population with two age classes: non-seniors (type 1) and seniors (type 2). We will represent $\mathcal{B}$ in the form $\mathcal{B} = b \circ \eta$, where b, η are 2×2 square matrices representing respectively the rate of transmission between age classes in the absence of restrictions and the 'interaction factors'

between different types, that model the effect of restrictions. The interaction factors should all be between 0 and 1. We will take b to be fixed, i.e.,

$$b = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

and we want to partially control η while maintaining criticality, i.e., $\alpha(\mathcal{Q}) = 0$. For simplicity we take η to be of the form

$$\eta = \begin{pmatrix} \eta_1 & \eta_2 \\ \eta_2 & \eta_2 \end{pmatrix},$$

where η_2 is the interaction factor between type-2 with other types, which we control, and η_1 is the interaction factor of type 1 with itself, which responds in order to maintain $\alpha(\mathcal{Q}) = 0$. We compute η_1 using the formula $\det(\mathcal{Q}) = \det \begin{pmatrix} \eta_1 b_{11}\mathcal{S}_1 - \gamma_1 & \eta_2 b_{12}\mathcal{S}_2 \\ \eta_2 b_{21}\mathcal{S}_1 & \eta_2 b_{22}\mathcal{S}_2 - \gamma_2 \end{pmatrix} = 0$, where $\mathcal{Q} = \mathcal{B}\mathcal{S} - \gamma$ and $\mathcal{B} = b \circ \eta$, obtaining

$$\eta_1 = \frac{\zeta + \gamma_1}{b_{11}\mathcal{S}(n)_1}, \text{ where } \zeta = \frac{b_{12}b_{21}\mathcal{S}(n)_1\mathcal{S}(n)_2(\eta_2)^2}{\eta_2 b_{22}\mathcal{S}(n)_2 - \gamma_2}.$$

Note that $\det(\mathcal{Q}) = 0$ does not imply $\alpha(\mathcal{Q}) = 0$, however, we expect that the two should correspond as long as $\alpha(\mathcal{Q}) < 0$ when $\eta_1 = 0$ and $\alpha(\mathcal{Q}) > 0$ when $\eta_1 = 1$, which we ensured was true throughout our simulations. The above are reasonable assumptions since if the first one fails then the infection will spread no matter how restricted type 1 is, and if the second one fails, the infection will die out no matter how unrestricted type 1 is. For all simulations, we chose

$$b = \begin{pmatrix} 4.221 & 0.9 \\ 2.1 & 1.224 \end{pmatrix}, \gamma = \begin{pmatrix} 0.4 & 0.2 \end{pmatrix}, \epsilon = 0.01,$$
$$\mathcal{R}_0 = \begin{pmatrix} 0 & 0 \end{pmatrix} \text{ and } u_0 = \begin{pmatrix} 0.6 & 0.4 \end{pmatrix}.$$

This gives $\mathcal{I}_0 = \begin{pmatrix} 0.006 & 0.004 \end{pmatrix}$. Finally, taking $\mathcal{S}_0 = \begin{pmatrix} 0.6 & 0.4 \end{pmatrix} - \mathcal{I}_0$ gives $\mathcal{S}_0 = \begin{pmatrix} 0.594 & 0.396 \end{pmatrix}$. In the first simulation, we chose $\eta_2 = 0.2$. The top plot of Figure 4.2 represents the susceptible population ($\mathcal{S}$) over time. The x-axis represents time, and the y-axis represents the proportion of the population that is susceptible. There are two lines in the plot: "$\mathcal{S}$-non-seniors " and "$\mathcal{S}$-seniors," which correspond to different age groups or subpopulations. The first plot shows how the proportion of susceptible individuals changes over time. The bottom plot shows the proportion of infected individuals get

recovered over time. The middle plot shows the infectious population ($\mathcal{I}$) over time. We notice a fast, immediate change as the $\mathcal{I}$ values settle onto a left Perron vector for $\mathcal{Q}(0)$. Then, as the number of susceptible gradually decreases, we notice a relative increase in non-senior infectious (type 1), due to relaxing restrictions among themselves.

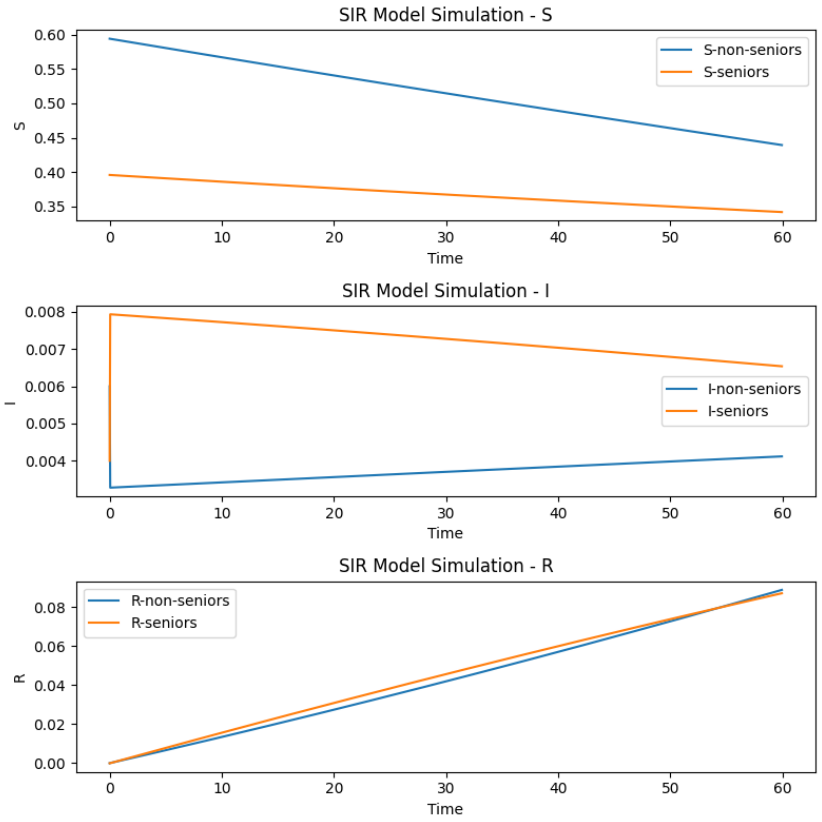


Figure 4.2: Structured SIR model with small $\mathcal{I}$ and a restriction on $\mathcal{B}$ (taking $\eta_2 = 0.2$) such that $\alpha(\mathcal{Q}) = 0$.

We explore different interaction factors for a certain time period to see their impact on disease behaviour. The idea is to reduce the interaction value η_2 to 10% from the regular interaction for the first 15 days then increase the value to 25% to see the changes for $15 - 30$ days. We did the changes back and forth for $30 - 45$ and $45 - 60$ days. Figure 4.3 depicts the result.

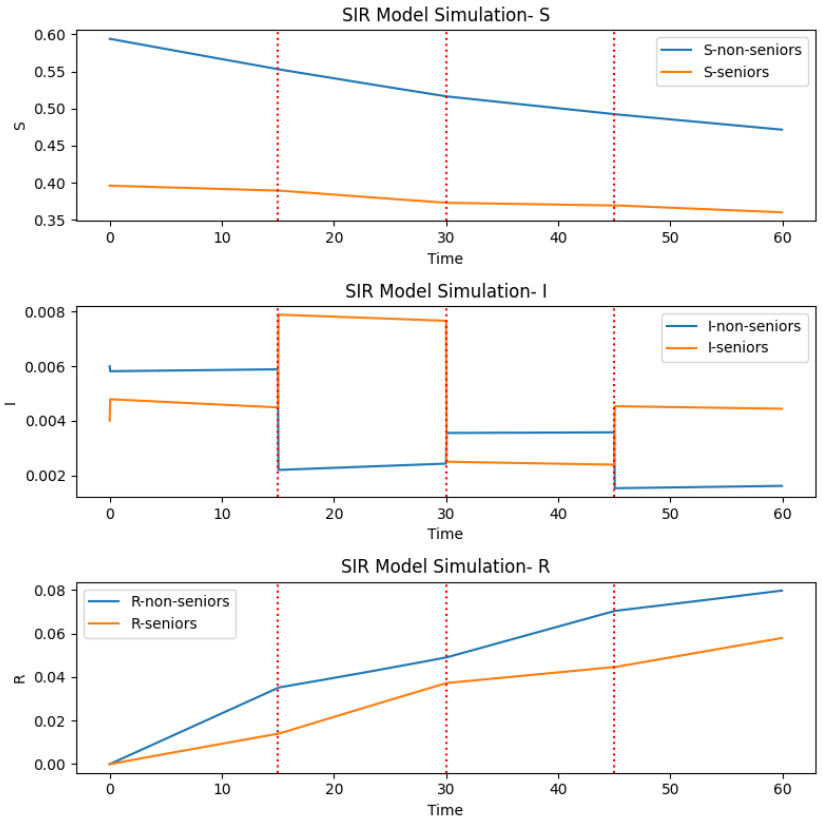


Figure 4.3: Structured SIR model with different η_2 values on the given time intervals. For times $0-15 : \eta_2 = 0.1$, $15-30 : \eta_2 = 0.25$, $30-45 : \eta_2 = 0.1$, $45-60 : \eta_2 = 0.25$.

In this example, during the initial 15 days, the primary intervention method is to reduce interactions between seniors and the rest of the population. By examining the upper plot (susceptibles), we can observe a gradual and consistent decrease in the susceptible population ($\mathcal{S}$) due to the implemented restriction measures. Every 15 days, slight discontinuous changes in the slopes of the curves can be noticed, which can be attributed to the sudden change in the parameter η_2, and similarly in the bottom plot (recovered). The middle plot describes the $\mathcal{I}$ dynamics. When η_2 is set to 0.1, both the non-seniors and seniors infected populations remain relatively stable for a duration of 15 days. However, when η_2 is increased to 0.25, the number of infectious seniors increases (due to increased interaction) and the number of

infectious non-seniors decreases (η_1 decreases to maintain $\alpha(\mathcal{Q}) = 0$). These changes occur suddenly. If this was simulated on the original time scale with a fairly small ϵ, at each time of change, there would be a relatively quick exponential decay towards the new 'quasi-equilibrium' values. The pattern is analogous at days 30 and 45.

Overall, we see from these results that when control measures change suddenly, the $\mathcal{I}$ value adjusts discontinuously to the new 'quasi-equilibrium' determined by $\mathcal{Q}$ at the moment of change. Between these times, the gradual movement of $\mathcal{I}$ values is due to η_1 increasing in response to gradually decreasing $\mathcal{S}$ values. In other words, the response variable adjusts to maintain criticality. Primarily we discussed here two age classes but for a large number of age classes, we can also use the same algorithm to analyze the slow-fast behaviour.

Chapter 5

Conclusion

In this chapter, I give a brief summary of my thesis, highlighting some of the implications of our results and future work in applying them.

After introducing the Perron value and Perron vectors r, u, v of a primitive matrix A, we analyzed the effect on r, u, v of perturbations in A. This led to the use of the group inverse, for which we obtained some useful formulas involving the stochasticized Markov chain with transition matrix $D^{-1}(A/r)D$ where D is the diagonal matrix with entries $D_{ii} = v_i$. We inserted these formulas into the perturbation equation in order to understand the factors (relating to the Markov chain) that determine the change in u and v. We found for example that v_i increases in response to an increase along the edge (k, ℓ) if and only if the Markov chain reaches k on average more quickly when it is initialized at i than when it is initialized to the stationary distribution. As a demonstration, we used these ideas to obtain an intuitive and simple proof of a result from [38] concerning the spread of infection over a star network.

In our analysis, we examine the $Q^{\#}$ formula for both the primitive and the stochastic matrix case, and in both the discrete and continuous time settings. Our alternative interpretation of the $Q^{\#}$ formula in the context of continuous-time Markov chains provides a fresh perspective and enhances its comprehensibility. By employing the concept of hitting time, we gain a tangible understanding of the connections that promote or inhibit the spread of infections. This understanding may serve as a basis for designing optimal contact networks that minimize the risk of disease transmission, subject to various constraints.

An important object from our perturbation analysis, the canonical flow of Perron vectors u, v with respect to a matrix-valued function A (or directly with respect to Q), makes it possible to understand changes in Perron vectors in terms of an associated Markov chain. In Chapter 4 we find that the canonical flow equations show up naturally in the context of infection spread near

the disease-free set of a critical structured SIR model. Our findings demonstrate that, under these assumptions, the $\mathcal{I}$ variable follows the left part (u variable) of the canonical flow equations, driven by $Q = \beta \circ S - \mathrm{diag}(\gamma)$ where β, γ are transmission and recovery rates and S is the vector giving the number of susceptibles of each type.

Overall, the results presented in this thesis underscore the effectiveness of perturbation methods, particularly the canonical flow equations, in analyzing the dynamics of infection spread and establishing crucial relationships between key variables. By interpreting these equations in terms of hitting times, we gain comprehensive insights into the underlying processes, paving the way for the development of effective strategies to control and manage the spread of infectious diseases.

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